

Sid Mathur's Thematic ETF Screener – Daily Report

2026-06-30

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Action Center

Run date: 2026-06-30 **Universe:** 660 ETFs / 26 clusters

Full PC1: FLAT **Clusters long:** 8/26 **ETFs long:** 235/660 (36%)

Today's events (cluster level)

spread_buy = 5dma – 50dma; spread_sell = 5dma – 20dma (price units). A BUY fired today iff spread_buy crossed above 0; a SELL iff spread_sell crossed below the cushioned threshold.

SELL (1)

cluster	theme	spread_sell	p_h5	nss
cluster_7	Emerging Markets Equity	-2.094	0.94	-0.794

Imminent (cluster level)

Ranked by short-horizon NSS (h=1 day) so candidates that could trigger *tomorrow* surface above those that need several days; p_h5 is the 5-day probability for context. **spread_buy = 5dma – 50dma, spread_sell = 5dma – 20dma** (price units; closer to 0 = closer to crossover). **Imminent BUYs are restricted to clusters whose 252d trend filter currently approves** – otherwise the cross would be blocked at entry, so the imminence is moot.

Imminent BUY (top 5 by h=1 NSS, filter-approved)

cluster	theme	spread_buy	p_h1	p_h5	nss_h1	nss_h5
cluster_1	U.S. Mid Cap Equity	5.302	1.000	1.00	1.000	1.00
cluster_14	Financials	3.277	1.000	1.00	1.000	1.00
cluster_24	Cash / Ultra-Short Fixed Income	0.116	1.000	0.81	1.000	0.76
cluster_15	Real Estate	2.262	1.000	0.95	1.000	0.90
cluster_22	Japan Equity	3.099	0.995	0.76	0.183	-0.01

Imminent SELL (top 2 by h=1 NSS)

cluster	theme	spread_sell	p_h1	p_h5	nss_h1	nss_h5
cluster_19	Broad Commodities	-10.596	1	1	-1	-1
cluster_25	International Fixed Income	-0.770	1	1	-1	-1

Full Sample PC1

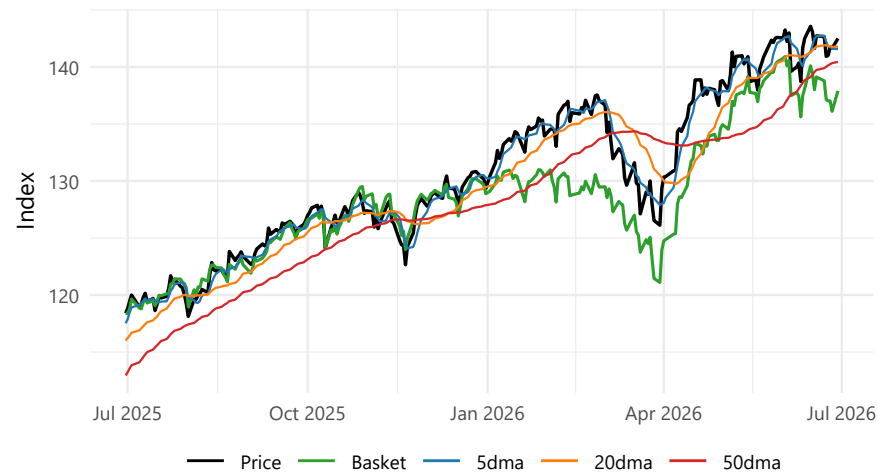


Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

Replication basket: VOO basket cor: 0.94 sticky

Current state: FLAT

5dma: 141.561 20dma: 141.763 50dma: 140.441

Spread vs 50dma: +1.120 Spread vs 20dma: -0.202

P(buy crossover within 5 days): 47.6% P(sell): 80.8%

Position Sheet

Replicate each currently-long cluster via its 1-5 member basket. Weights are NNLS (non-negative least squares) – long-only and tradeable – and sum to ~1 per cluster for unit-variance replication. Scale each cluster's basket to your target cluster allocation.

Active positions (long clusters)

ticker	cluster	theme	nss	days	return	pos_wt	cor
TLT	cluster_13	Fixed Income	1.000	9	1.3%	13.88%	0.97
VHT	cluster_11	Healthcare	1.000	36	11.1%	9.72%	0.93
MUB	cluster_16	Municipal Bond	0.996	21	0.6%	18.09%	0.95
BND	cluster_6	Aggregate Fixed Income	0.932	2	0.1%	14.58%	0.99
XLP	cluster_18	Consumer Staples	0.906	1	-0.4%	6.82%	0.99
XLU	cluster_18	Consumer Staples	0.906	1	-0.4%	4.22%	0.99
VTV	cluster_2	U.S. Large Cap Value	0.819	53	5.5%	16.16%	0.98
BWX	cluster_25	International Fixed Income	-1.000	20	-2.1%	15.01%	0.93
PDBC	cluster_19	Broad Commodities	-1.000	17	-14.2%	5.97%	0.98

Watchlist (imminent-buy clusters, $p > 80\%$, not yet long)

ticker	cluster	theme	p_h5	nss	weight	cor
XLF	cluster_14	Financials	1.00	0.997	1.096	0.96
IJH	cluster_1	U.S. Mid Cap Equity	1.00	1.000	1.016	0.99
VNQ	cluster_15	Real Estate	0.95	0.901	0.970	0.99
ICSH	cluster_24	Cash / Ultra-Short Fixed In...	0.81	0.760	0.413	0.84
SHV	cluster_24	Cash / Ultra-Short Fixed In...	0.81	0.760	0.297	0.84
MINT	cluster_24	Cash / Ultra-Short Fixed In...	0.81	0.760	0.177	0.84
GBIL	cluster_24	Cash / Ultra-Short Fixed In...	0.81	0.760	0.121	0.84

Cluster Overview

name is derived from the basket members' categories (e.g. "Technology" or "Tech + Financials"). change shows the LONG/FLAT shift vs the prior run: →LONG or →FLAT for a state flip, new if this cluster had no match in yesterday's bundle, shifted if the cluster's identity changed (top-3 members by |PC1 weight| no longer match), empty if state is unchanged. dNSS is today's NSS minus the prior run's.

	id	name	category	n	signal	change	days	return	nss	dNSS
1	cluster_1	U.S. Mid Cap Equity	U.S. Mid Cap Equi...	94	FLAT	shifted			1.000	-0.000
11	cluster_11	Healthcare	Healthcare	17	LONG	shifted	36	11.1%	1.000	+1.837
13	cluster_13	Fixed Income	Fixed Income	15	LONG	shifted	9	1.3%	1.000	+1.997
14	cluster_14	Financials	Financials	14	FLAT				0.997	-0.002
16	cluster_16	Municipal Bond	Municipal Bond	13	LONG		21	0.6%	0.996	+0.016
6	cluster_6	Aggregate Fixed Income	Aggregate Fixed I...	43	LONG	shifted	2	0.1%	0.932	+1.024
18	cluster_18	Consumer Staples	Consumer Staples	11	LONG		1	-0.4%	0.906	+0.231
15	cluster_15	Real Estate	Real Estate	14	FLAT				0.901	+0.174
2	cluster_2	U.S. Large Cap Value	U.S. Large Cap Va...	75	LONG	shifted	53	5.5%	0.819	+0.310
24	cluster_24	Cash / Ultra-Short Fixed In...	Cash / Ultra-Shor...	9	FLAT				0.760	+0.245
26	cluster_26	Leveraged / Inverse	Leveraged / Inver...	2	FLAT	shifted			0.339	+0.200
22	cluster_22	Japan Equity	Japan Equity	9	FLAT				-0.013	-0.526
20	cluster_20	India Equity	India Equity	10	FLAT				-0.075	-0.027
21	cluster_21	Inflation-Protected Fixed I...	Inflation-Protect...	10	FLAT				-0.650	+0.268
3	cluster_3	U.S. Large Cap Growth	U.S. Large Cap Gr...	65	FLAT	shifted			-0.652	+0.296
7	cluster_7	Emerging Markets Equity	Emerging Markets ...	25	FLAT	shifted			-0.794	-0.561
10	cluster_10	High Yield Bond	High Yield Bond	21	FLAT	shifted			-0.869	+0.131
8	cluster_8	Asia-Pacific Equity	Asia-Pacific Equi...	24	FLAT	shifted			-0.979	+0.011
4	cluster_4	Developed ex-US Equity	Developed ex-US E...	60	FLAT	shifted			-0.985	-1.961
5	cluster_5	U.S. Large Cap Equity	U.S. Large Cap Eq...	55	FLAT				-0.988	-0.006
12	cluster_12	China Equity	China Equity	15	FLAT	shifted			-0.999	-1.999
17	cluster_17	Gold	Gold	12	FLAT				-0.999	-0.001
23	cluster_23	Bank Loan / Floating Rate	Bank Loan / Float...	9	FLAT				-0.999	-0.002
9	cluster_9	Materials	Materials	21	FLAT	shifted			-1.000	-0.603
19	cluster_19	Broad Commodities	Broad Commodities	11	LONG		17	-14.2%	-1.000	+0.000
25	cluster_25	International Fixed Income	International Fix...	6	LONG		20	-2.1%	-1.000	+0.000

Cluster Details

Cluster 1 – U.S. Mid Cap Equity

Members: 94 **Var explained:** 0.861 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +41.9%)
- Latest sell_cross: 2026-05-18
- Current 252d return: +28.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-18 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Industrials	15	0.1503
U.S. Small Cap Equity	10	0.1121
U.S. Small Cap Growth	6	0.0664
U.S. Small Cap Value	6	0.0661
U.S. Mid Cap Growth	6	0.0649
Consumer Discretionary	6	0.0589
U.S. Mid Cap Equity	5	0.0566
U.S. Mid Cap Value	3	0.0334

Utilities	3	0.0296
U.S. Momentum Equity	3	0.0287
U.S. Small Cap Momentum Equity	2	0.0212
Leveraged / Inverse	2	0.0211
Growth Equity	2	0.0209
Momentum Equity	2	0.0200
Unknown	1	0.0114
U.S. Small Cap ESG Equity	1	0.0114
U.S. Mid/Small Cap Equity	1	0.0113
U.S. Small Cap Multifactor Equity	1	0.0113
U.S. Small Cap Fundamental Equity	1	0.0113
U.S. Small/Mid Cap Fundamental Equity	1	0.0113
Mid Cap Value Equities	1	0.0112
U.S. Extended Market Equity	1	0.0111
U.S. ESG Equity	1	0.0111
Dividend Equity	1	0.0110
U.S. Mid Cap Quality Equity	1	0.0110
U.S. Small/Mid Cap Equity	1	0.0109
Small Cap Value Equities	1	0.0109
U.S. Mid Cap Dividend Equity	1	0.0109
U.S. Equity	1	0.0108
U.S. Small Cap Industrials	1	0.0107
U.S. Small Cap Dividend Equity	1	0.0106
Value Equity	1	0.0106
U.S. Mid Cap Momentum Equity	1	0.0105
U.S. Micro Cap Equity	1	0.0105
U.S. Small Cap Consumer Discretionary	1	0.0101
Technology	1	0.0101
IPO Equity	1	0.0096

Replication basket – IJH (cor=0.99) basket lead at lag 1: -0.008

ticker	rep_wt	name	category	etf_signal
IJH	1.016	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
27	NUSC	0.0114	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG Equity	FLAT
31	MDY	0.0114	State Street SPDR S&P MIDCAP 400 ETF Tr...	U.S. Mid Cap Equity	FLAT
39	IVOO	0.0114	Vanguard S&P Mid-Cap 400 ETF		FLAT

46	IJH	0.0114	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
47	SPMD	0.0114	State Street SPDR Portfolio S&P 400 Mid...	U.S. Mid Cap Equity	FLAT
49	VB	0.0114	Vanguard Small-Cap ETF	U.S. Small Cap Equity	FLAT
22	SCHM	0.0113	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	LONG
23	FNX	0.0113	First Trust Mid Cap Core AlphaDEX Fund	U.S. Mid/Small Cap Equity	FLAT
28	PRFZ	0.0113	Invesco RAFI US 1500 Small-Mid ETF	U.S. Small/Mid Cap Funda...	FLAT
29	SMLF	0.0113	iShares U.S. Small-Cap Equity Factor ETF	U.S. Small Cap Multifact...	FLAT
50	SCHA	0.0113	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
67	FNDA	0.0113	Schwab Fundamental U.S. Small Company E...	U.S. Small Cap Fundament...	FLAT
76	VBR	0.0113	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	FLAT
4	IWM	0.0112	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	FLAT
32	IJT	0.0112	iShares S&P SmallCap 600 Growth ETF	U.S. Small Cap Growth	FLAT
33	IJK	0.0112	iShares S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
34	VIOG	0.0112	Vanguard S&P Small-Cap 600 Growth ETF	U.S. Small Cap Growth	FLAT
35	IVOG	0.0112	Vanguard S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
36	MDYG	0.0112	State Street SPDR S&P 400 Mid Cap Growt...	U.S. Mid Cap Growth	LONG
37	SLYG	0.0112	State Street SPDR S&P 600 Small Cap Gro...	U.S. Small Cap Growth	FLAT
40	VIOO	0.0112	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equity	FLAT
41	VTWO	0.0112	Vanguard Russell 2000 ETF	U.S. Small Cap Equity	FLAT
43	IJR	0.0112	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	FLAT
48	SPSM	0.0112	State Street SPDR Portfolio S&P 600 Sma...	U.S. Small Cap Equity	FLAT
62	EZM	0.0112	WisdomTree U.S. MidCap Fund	Mid Cap Value Equities	FLAT
72	MDYV	0.0112	State Street SPDR S&P 400 Mid Cap Value...	U.S. Mid Cap Value	FLAT
24	FYX	0.0111	First Trust Small Cap Core AlphaDEX Fund	U.S. Small Cap Equity	FLAT
45	VXF	0.0111	Vanguard Extended Market ETF	U.S. Extended Market Equ...	FLAT
63	PSC	0.0111	Principal U.S. Small-Cap ETF	U.S. Small Cap Equity	FLAT
68	IWN	0.0111	iShares Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
69	IJJ	0.0111	iShares S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
71	IVOV	0.0111	Vanguard S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
78	ETHO	0.0111	Amplify Etho Climate Leadership U.S. ETF	U.S. ESG Equity	FLAT
86	RWK	0.0111	Invesco S&P MidCap 400 Revenue ETF	U.S. Mid Cap Equity	FLAT
15	FXR	0.0110	First Trust Industrials AlphaDEX	Industrials	LONG
44	ISCG	0.0110	iShares Morningstar Small-Cap Growth ETF	Growth Equity	FLAT
75	VTWV	0.0110	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
82	XMHQ	0.0110	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Quality Equ...	FLAT
93	SDVY	0.0110	First Trust SMID Cap Rising Dividend Ac...	Dividend Equity	LONG
30	IWO	0.0109	iShares Russell 2000 Growth Fund	U.S. Small Cap Growth	LONG
38	VTWG	0.0109	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Growth	LONG
42	VBK	0.0109	Vanguard Small-Cap Growth ETF	U.S. Small Cap Growth	FLAT
60	DON	0.0109	WisdomTree U.S. MidCap Dividend Fund	U.S. Mid Cap Dividend Eq...	FLAT
61	EES	0.0109	WisdomTree U.S. SmallCap Fund	Small Cap Value Equities	FLAT
70	IJS	0.0109	iShares S&P SmallCap 600 Value ETF	U.S. Small Cap Value	LONG
73	SLYV	0.0109	State Street SPDR S&P 600 Small Cap Val...	U.S. Small Cap Value	FLAT
74	VIOV	0.0109	Vanguard S&P Small-Cap 600 Value ETF	U.S. Small Cap Value	FLAT
79	RWJ	0.0109	Invesco S&P SmallCap 600 Revenue ETF	U.S. Small Cap Equity	FLAT

94	ISMD	0.0109	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap Equity	LONG
19	IYJ	0.0108	iShares U.S. Industrials ETF	Industrials	FLAT
20	VIS	0.0108	Vanguard Industrials ETF	Industrials	FLAT
21	FIDU	0.0108	Fidelity MSCI Industrials Index ETF	Industrials	FLAT
87	OMFS	0.0108	Invesco Russell 2000 Dynamic Multifacto...	U.S. Equity	FLAT
91	XSMO	0.0108	Invesco S&P SmallCap Momentum ETF	U.S. Small Cap Momentum ...	FLAT
14	IMCG	0.0107	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
80	PSCI	0.0107	Invesco S&P SmallCap Industrials ETF	U.S. Small Cap Industria...	FLAT
81	PAVE	0.0107	Global X U.S. Infrastructure Developmen...	Industrials	FLAT
85	RFV	0.0107	Invesco S&P MidCap 400 Pure Value ETF	Leveraged / Inverse	FLAT
51	FXD	0.0106	First Trust Cons. Discret. AlphaDEX	Consumer Discretionary	LONG
59	DES	0.0106	WisdomTree U.S. SmallCap Dividend Fund	U.S. Small Cap Dividend ...	FLAT
92	FYT	0.0106	First Trust Small Cap Value AlphaDEX Fu...	Value Equity	FLAT
10	XMMO	0.0105	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Momentum Eq...	LONG
26	IWC	0.0105	iShares Microcap ETF	U.S. Micro Cap Equity	FLAT
1	XLI	0.0104	State Street Industrial Select Sector S...	Industrials	FLAT
25	DWAS	0.0104	Invesco Dorsey Wright SmallCap Momentum...	U.S. Small Cap Momentum ...	FLAT
58	RSPD	0.0104	Invesco S&P 500 Equal Weight Consumer D...	Consumer Discretionary	LONG
83	VFMO	0.0104	Vanguard U.S. Momentum Factor ETF	U.S. Momentum Equity	LONG
88	RZV	0.0104	Invesco S&P SmallCap 600 Pure Value ETF	Leveraged / Inverse	FLAT
11	IWP	0.0103	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	LONG
13	VOT	0.0103	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
18	FIW	0.0103	First Trust Water ETF	Utilities	LONG
53	FTXR	0.0103	First Trust Nasdaq Transportation ETF	Industrials	LONG
84	PKB	0.0102	Invesco Building & Construction ETF	Industrials	LONG
6	FTC	0.0101	First Trust Large Cap Growth AlphaDEX F...	Technology	LONG
90	PSCD	0.0101	Invesco S&P SmallCap Consumer Discretio...	U.S. Small Cap Consumer ...	LONG
5	FV	0.0100	First Trust Dorsey Wright Focus 5 ETF	Momentum Equity	LONG
7	PDP	0.0100	Invesco Dorsey Wright Momentum ETF	Momentum Equity	LONG
16	PHO	0.0100	Invesco Water Resources ETF	Utilities	LONG
66	XTN	0.0100	State Street SPDR S&P Transportation ETF	Industrials	FLAT
9	RPG	0.0099	Invesco S&P 500 Pure Growth ETF	Growth Equity	LONG
89	PRN	0.0099	Invesco Dorsey Wright Industrials Momen...	Industrials	FLAT
2	XRT	0.0098	State Street SPDR S&P Retail ETF	Consumer Discretionary	LONG
52	PEJ	0.0098	Invesco Leisure and Entertainment ETF	Consumer Discretionary	LONG
57	IYT	0.0098	iShares U.S. Transportation ETF	Industrials	FLAT
8	FPX	0.0096	First Trust US Equity Opportunities ETF	IPO Equity	FLAT
65	XHB	0.0096	State Street SPDR S&P Homebuilders ETF	Consumer Discretionary	LONG
17	CGW	0.0093	Invesco S&P Global Water Index ETF	Utilities	LONG
77	QMOM	0.0093	Alpha Architect U.S. Quantitative Momen...	U.S. Momentum Equity	FLAT
64	XAR	0.0092	State Street SPDR S&P Aerospace & Defen...	Industrials	FLAT
54	PPA	0.0091	Invesco Aerospace & Defense ETF	Industrials	LONG
12	MTUM	0.0090	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	LONG
3	JETS	0.0089	U.S. Global Jets ETF	Industrials	FLAT
56	ITB	0.0087	iShares U.S. Home Construction ETF	Consumer Discretionary	LONG

55 ITA 0.0085 iShares U.S. Aerospace & Defense ETF Industrials LONG

Cluster 2 – U.S. Large Cap Value

Members: 75 **Var explained:** 0.878 **Signal:** LONG since 2026-04-13 (53 days, 5.5%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +22.7%)
- Latest sell_cross: 2026-03-06
- Current 252d return: +20.5% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



Category breakdown

category	n_members	abs_weight
U.S. Dividend Equity	22	0.2917
U.S. Large Cap Value	10	0.1398
Value Equity	4	0.0541
U.S. Mid Cap Equity	4	0.0538
U.S. Low Volatility Equity	4	0.0500

U.S. Multifactor Equity	3	0.0410
U.S. Value Factor Equity	3	0.0401
Dividend Equity	3	0.0388
U.S. Fundamental Equity	2	0.0279
U.S. Mid Cap Value	2	0.0278
U.S. Equity	2	0.0277
U.S. Equal Weight Equity	2	0.0277
U.S. Small Cap Low Volatility	2	0.0253
U.S. Mega Cap Value	1	0.0138
U.S. Revenue Weighted Equity	1	0.0138
Large Cap Blend Equities	1	0.0136
U.S. Size Factor Equity	1	0.0136
U.S. Buyback Equity	1	0.0134
U.S. Large Cap Equity	1	0.0133
U.S. Quality / Factor Equity	1	0.0130
U.S. Mid Cap Low Volatility	1	0.0130
U.S. Defensive Equity	1	0.0128
Global Equity	1	0.0122
Communication Services	1	0.0112
Real Estate	1	0.0108

Replication basket – VTV (cor=0.98) basket lead at lag 1: 0.005

ticker	rep_wt	name	category	etf_signal
VTV	1.003	Vanguard Value ETF	U.S. Large Cap Value	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
27	IWD	0.0141	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
3	RSP	0.0140	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity	LONG
12	PRF	0.0140	Invesco RAFI US 1000 ETF	U.S. Fundamental Equity	LONG
30	JPUS	0.0140	JPMorgan Diversified Return U.S. Equity...	U.S. Multifactor Equity	FLAT
35	VONV	0.0140	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	LONG
39	ILCV	0.0140	iShares Morningstar Large-Cap Value ETF	U.S. Large Cap Value	FLAT
41	IUSV	0.0140	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	FLAT
42	SCHV	0.0140	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	LONG
45	VTV	0.0140	Vanguard Value ETF	U.S. Large Cap Value	LONG
61	VOE	0.0140	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
66	DTD	0.0140	WisdomTree U.S. Total Dividend Fund	U.S. Dividend Equity	LONG
17	CFA	0.0139	VictoryShares US 500 Volatility Wtd ETF	U.S. Multifactor Equity	FLAT

19	NULV	0.0139	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	FLAT
23	FNDX	0.0139	Schwab Fundamental U.S. Large Company E...	U.S. Fundamental Equity	LONG
29	IVE	0.0139	iShares S&P 500 Value ETF	U.S. Large Cap Value	FLAT
33	VOOV	0.0139	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	LONG
38	VYM	0.0139	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	FLAT
43	SPYV	0.0139	State Street SPDR Portfolio S&P 500 Val...	U.S. Large Cap Value	FLAT
71	DGRO	0.0139	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG
13	RWL	0.0138	Invesco S&P 500 Revenue ETF	U.S. Revenue Weighted Eq...	LONG
15	AIVL	0.0138	WisdomTree U.S. AI Enhanced Value Fund	U.S. Value Factor Equity	LONG
25	IWX	0.0138	iShares Russell Top 200 Value ETF	Value Equity	LONG
26	ONEV	0.0138	State Street SPDR Russell 1000 Low Vola...	U.S. Equity	FLAT
36	MGV	0.0138	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	LONG
46	FTA	0.0138	First Trust Large Cap Value AlphaDEX Fu...	Value Equity	LONG
57	IWS	0.0138	iShares Russell Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
58	ONEY	0.0138	State Street SPDR Russell 1000 Yield Fo...	U.S. Dividend Equity	FLAT
65	DLN	0.0138	WisdomTree U.S. LargeCap Dividend Fund	U.S. Dividend Equity	LONG
74	JPME	0.0138	JPMorgan Diversified Return U.S. Mid Ca...	U.S. Equity	FLAT
24	EQAL	0.0137	Invesco Russell 1000 Equal Weight ETF	U.S. Equal Weight Equity	FLAT
63	PFM	0.0137	Invesco Dividend Achievers ETF	U.S. Dividend Equity	FLAT
6	FEX	0.0136	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities	LONG
20	SDY	0.0136	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	LONG
31	SIZE	0.0136	iShares MSCI USA Size Factor ETF	U.S. Size Factor Equity	LONG
10	JHMM	0.0135	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity	FLAT
21	FDVV	0.0135	Fidelity High Dividend ETF	U.S. Dividend Equity	FLAT
37	SCHD	0.0135	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG
40	IMCB	0.0135	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	LONG
51	DVY	0.0135	iShares Select Dividend ETF	U.S. Dividend Equity	FLAT
72	VIG	0.0135	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
75	JVAL	0.0135	JPMorgan U.S. Value Factor ETF	U.S. Value Factor Equity	LONG
4	FVD	0.0134	First Trust VL Dividend	U.S. Dividend Equity	LONG
5	PKW	0.0134	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	LONG
7	PWV	0.0134	Invesco Large Cap Value ETF	Value Equity	LONG
28	IWR	0.0134	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	LONG
44	VO	0.0134	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	LONG
49	SDOG	0.0134	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	FLAT
68	DIA	0.0133	State Street SPDR Dow Jones Industrial ...	U.S. Large Cap Equity	LONG
8	RDVY	0.0132	First Trust Rising Dividend Achievers E...	U.S. Dividend Equity	FLAT
60	SPYD	0.0132	State Street SPDR Portfolio S&P 500 Hig...	U.S. Dividend Equity	FLAT
70	LGLV	0.0132	State Street SPDR US Large Cap Low Vola...	U.S. Low Volatility Equi...	LONG
11	REGL	0.0131	ProShares S&P MidCap 400 Dividend Arist...	Dividend Equity	LONG
53	RPV	0.0131	Invesco S&P 500 Pure Value ETF	Value Equity	FLAT
64	OUSL	0.0131	ALPS O'Shares U.S. Quality Dividend ETF	Dividend Equity	FLAT
14	DHS	0.0130	WisdomTree U.S. High Dividend Fund	U.S. Dividend Equity	FLAT
18	CFO	0.0130	VictoryShares US 500 Enhanced Volatilit...	U.S. Multifactor Equity	FLAT
55	XMLV	0.0130	Invesco S&P MidCap Low Volatility ETF	U.S. Mid Cap Low Volatil...	LONG

59	SMMV	0.0130	iShares MSCI USA Small-Cap Min Vol Fact...	U.S. Small Cap Low Volat...	LONG
62	FTCS	0.0130	First Trust Capital Strength ETF	U.S. Quality / Factor Eq...	LONG
32	VLUE	0.0129	iShares MSCI USA Value Factor ETF	U.S. Value Factor Equity	LONG
16	CDC	0.0128	VictoryShares US EQ Income Enhanced Vol...	U.S. Defensive Equity	FLAT
52	RDIV	0.0128	Invesco S&P Ultra Dividend Revenue ETF	U.S. Dividend Equity	FLAT
69	USMV	0.0128	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility Equi...	LONG
47	PEY	0.0126	Invesco High Yield Equity Dividend Achi...	U.S. Dividend Equity	FLAT
54	SPHD	0.0126	Invesco S&P 500 High Dividend Low Volat...	Dividend Equity	LONG
9	FDL	0.0125	First Trust Morningstar ETF	U.S. Dividend Equity	FLAT
48	DIV	0.0124	Global X Super Dividend ETF	U.S. Dividend Equity	LONG
56	XSLV	0.0123	Invesco S&P SmallCap Low Volatility ETF	U.S. Small Cap Low Volat...	FLAT
2	ACWV	0.0122	iShares MSCI Global Min Vol Factor ETF	Global Equity	FLAT
22	LVHD	0.0122	Franklin U.S. Low Volatility High Divid...	U.S. Low Volatility Equi...	LONG
34	HDV	0.0122	iShares Core High Dividend ETF	U.S. Dividend Equity	LONG
50	SMDV	0.0122	ProShares Russell 2000 Dividend Growers...	U.S. Dividend Equity	FLAT
67	SPLV	0.0118	Invesco S&P 500 Low Volatility ETF	U.S. Low Volatility Equi...	LONG
73	IYZ	0.0112	iShares U.S. Telecommunications ETF	Communication Services	FLAT
1	REM	0.0108	iShares Mortgage Real Estate ETF	Real Estate	FLAT

Cluster 3 – U.S. Large Cap Growth

Members: 65 **Var explained:** 0.817 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +34.4%)
- Latest sell_cross: 2026-06-26
- Current 252d return: +30.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-26 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Technology	34	0.5233
U.S. Large Cap Growth	11	0.1820
U.S. Equity	2	0.0327
Growth Equity	2	0.0325
Convertible Bond	2	0.0301
Financials	2	0.0290
Consumer Discretionary	2	0.0281
U.S. Mega Cap Growth	1	0.0164
U.S. Equal Weight Equity	1	0.0164
U.S. Small Cap Technology	1	0.0150
IPO Equity	1	0.0150
Options Income	1	0.0149
Trend-Following Equity	1	0.0145
International IPO Equity	1	0.0136
U.S. Trend-Following Equity	1	0.0127
Israel Equity	1	0.0125
Singapore Equity	1	0.0114

Replication basket – QQQ (cor=0.97) basket lead at lag 1: -0.017

ticker	rep_wt	name	category	etf_signal
QQQ	1.01	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
15	QQQ	0.0167	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	FLAT
38	IGM	0.0167	iShares Expanded Tech Sector ETF	Technology	FLAT
43	ONEQ	0.0167	Fidelity Nasdaq Composite Index ETF	U.S. Equity	FLAT
53	ILCG	0.0167	iShares Morningstar Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
31	XNTK	0.0166	State Street SPDR NYSE Technology ETF	Technology	LONG
40	IYW	0.0166	iShares U.S. Technology ETF	Technology	FLAT
42	NULG	0.0166	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
48	VGT	0.0166	Vanguard Information Tech ETF	Technology	FLAT
50	FTEC	0.0166	Fidelity MSCI Information Technology In...	Technology	FLAT
57	VUG	0.0166	Vanguard Growth ETF	U.S. Large Cap Growth	FLAT
8	AIQ	0.0165	Global X Artificial Intelligence & Tech...	Technology	FLAT
20	QTEC	0.0165	First Trust NASDAQ-100-Technology Secto...	Technology	LONG
26	FXL	0.0165	First Trust Technology AlphaDEX	Technology	FLAT
46	IWF	0.0165	iShares Russell 1000 Growth Fund	U.S. Large Cap Growth	FLAT
47	IVW	0.0165	iShares S&P 500 Growth ETF	U.S. Large Cap Growth	FLAT
49	VOOG	0.0165	Vanguard S&P 500 Growth ETF	U.S. Large Cap Growth	FLAT
51	VONG	0.0165	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Growth	FLAT
54	IUSG	0.0165	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Growth	FLAT
55	SCHG	0.0165	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
56	SPYG	0.0165	State Street SPDR Portfolio S&P 500 Gro...	U.S. Large Cap Growth	FLAT
19	QQEW	0.0164	First Trust Nasdaq-100 Select Equal Wei...	U.S. Equal Weight Equity	LONG
52	MGK	0.0164	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	FLAT
1	XLK	0.0163	State Street Technology Select Sector S...	Technology	FLAT
23	XT	0.0163	iShares Future Exponential Technologies...	Technology	FLAT
39	IXN	0.0163	iShares Global Tech ETF	Technology	FLAT
45	IWY	0.0163	iShares Russell Top 200 Growth ETF	Growth Equity	FLAT
37	PWB	0.0162	Invesco Large Cap Growth ETF	Growth Equity	FLAT
58	SPHB	0.0160	Invesco S&P 500 High Beta ETF	U.S. Equity	LONG
7	ROBO	0.0157	ROBO Global Robotics and Automation Ind...	Technology	FLAT
22	TDIV	0.0157	First Trust NASDAQ Technology Dividend ...	Technology	FLAT
6	BOTZ	0.0156	Global X Robotics & Artificial Intellig...	Technology	FLAT
65	QTUM	0.0156	Defiance Quantum ETF	Technology	LONG
30	FDN	0.0155	First Trust DJ Internet Index Fund	Technology	FLAT
3	SMH	0.0153	VanEck Semiconductor ETF	Technology	LONG
29	PNQI	0.0153	Invesco Nasdaq Internet ETF	Technology	FLAT
2	SOXX	0.0152	iShares PHLX SOX Semiconductor Sector I...	Technology	LONG

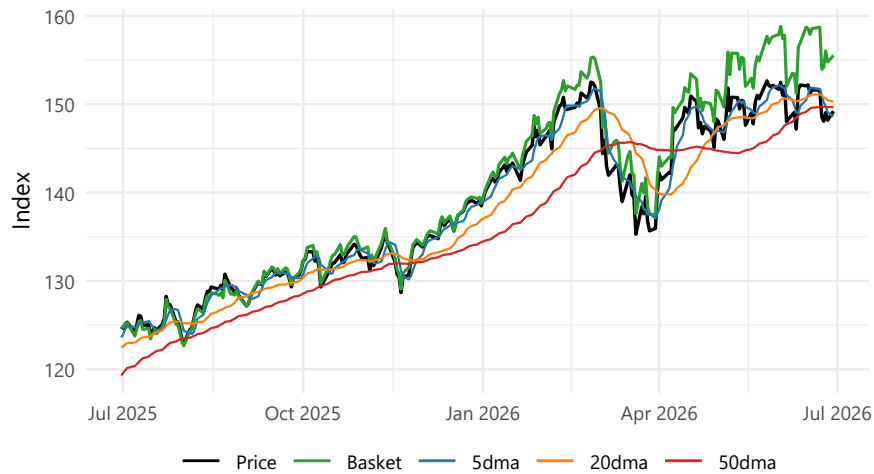
5	SKYY	0.0152	First Trust Cloud Computing ETF	Technology	FLAT
13	ARKQ	0.0152	ARK Autonomous Technology & Robotics ETF	Technology	LONG
41	CWB	0.0152	State Street SPDR Bloomberg Convertible...	Convertible Bond	FLAT
64	PTF	0.0152	Invesco Dorsey Wright Technology Moment...	Technology	LONG
17	NXTG	0.0151	First Trust Indxx NextG ETF	Technology	FLAT
24	XSD	0.0151	State Street SPDR S&P Semiconductor ETF	Technology	LONG
28	IPO	0.0150	Renaissance IPO ETF	IPO Equity	LONG
61	PSCT	0.0150	Invesco S&P SmallCap Information Techno...	U.S. Small Cap Technology	LONG
4	IGV	0.0149	iShares Expanded Tech-Software Sector E...	Technology	FLAT
21	PSI	0.0149	Invesco Semiconductors ETF	Technology	LONG
34	FINX	0.0149	Global X FinTech ETF	Financials	FLAT
36	QYLD	0.0149	Global X NASDAQ 100 Covered Call ETF	Options Income	FLAT
12	ARKW	0.0148	ARK Next Generation Internet ETF	Technology	FLAT
44	ICVT	0.0148	iShares Convertible Bond ETF	Convertible Bond	FLAT
63	FTXL	0.0148	First Trust Nasdaq Semiconductor ETF	Technology	LONG
11	ARKK	0.0146	ARK Innovation ETF	Technology	LONG
35	PTNQ	0.0145	Pacer Trendpilot 100 ETF	Trend-Following Equity	FLAT
25	IBUY	0.0143	Amplify Online Retail ETF	Consumer Discretionary	LONG
27	CIBR	0.0143	First Trust NASDAQ Cybersecurity ETF	Technology	FLAT
10	HACK	0.0141	Amplify Cybersecurity ETF	Technology	FLAT
18	PRNT	0.0141	3D Printing (The) ETF	Technology	FLAT
32	IPAY	0.0141	Amplify Digital Payments ETF	Financials	FLAT
14	ESPO	0.0137	VanEck Video Gaming and eSports ETF	Consumer Discretionary	FLAT
33	FPXI	0.0136	First Trust International Equity Opport...	International IPO Equity	FLAT
9	BLOK	0.0135	Amplify Blockchain Technology ETF	Technology	FLAT
59	PTLC	0.0127	Pacer Trendpilot US Large Cap ETF	U.S. Trend-Following Equ...	FLAT
60	EIS	0.0125	iShares MSCI Israel ETF	Israel Equity	FLAT
62	BLCN	0.0119	Siren NexGen Economy ETF	Technology	FLAT
16	EWS	0.0114	iShares MSCI Singapore ETF	Singapore Equity	LONG

Cluster 4 – Developed ex-US Equity

Members: 60 Var explained: 0.899 Signal: FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +47.8%)
- Latest sell_cross: 2026-06-25
- Current 252d return: +20.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-25 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
International Equity	19	0.3205
Developed ex-US Equity	9	0.1551
Global ex-US Equity	5	0.0861
Europe Equity	4	0.0684
International Dividend Equity	4	0.0669
Alternatives	4	0.0625
Eurozone Equity	3	0.0497
Commodity	1	0.0174
Global ex-US Small Cap Equity	1	0.0168
Germany Equity	1	0.0164
France Equity	1	0.0164
Italy Equity	1	0.0161
United Kingdom Equity	1	0.0160
Europe Financials	1	0.0160
Netherlands Equity	1	0.0159
Sweden Equity	1	0.0158
Spain Equity	1	0.0152
Austria Equity	1	0.0150
Poland Equity	1	0.0137

Replication basket – VEA (cor=0.99) basket lead at lag 1: -0.017

ticker	rep_wt	name	category	etf_signal
VEA	0.964	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	EFA	0.0175	iShares MSCI EAFE ETF	Developed ex-US Equity	FLAT
3	VEA	0.0175	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	FLAT
41	ESGD	0.0175	iShares ESG Aware MSCI EAFE ETF	International Equity	FLAT
51	IEFA	0.0175	iShares Core MSCI EAFE ETF	Developed ex-US Equity	FLAT
40	GSIE	0.0174	Goldman Sachs ActiveBeta International ...	Commodity	FLAT
52	SCHF	0.0174	Schwab International Equity ETF	Developed ex-US Equity	FLAT
53	SPDW	0.0174	State Street SPDR Portfolio Developed W...	Developed ex-US Equity	FLAT
16	DWM	0.0173	WisdomTree International Equity Fund	International Dividend E...	FLAT
25	TLTD	0.0173	FlexShares Morningstar Developed Market...	International Equity	FLAT
32	INTF	0.0173	iShares International Equity Factor ETF	International Equity	FLAT
47	IXUS	0.0173	iShares Core MSCI Total International S...	Global ex-US Equity	FLAT
49	VGK	0.0173	Vanguard FTSEEuropean ETF	Europe Equity	FLAT
6	IEV	0.0172	iShares Europe ETF	Europe Equity	FLAT
19	PXF	0.0172	Invesco RAFI Developed Markets ex-U.S. ...	International Equity	FLAT
29	ACWX	0.0172	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	FLAT
30	CWI	0.0172	State Street SPDR MSCI ACWI ex-US ETF	Global ex-US Equity	FLAT
33	IQLT	0.0172	iShares MSCI Intl Quality Factor ETF	International Equity	LONG
39	FNDF	0.0172	Schwab Fundamental International Equity...	International Equity	FLAT
46	IEUR	0.0172	iShares Core MSCI Europe ETF	Europe Equity	FLAT
48	VEU	0.0172	Vanguard FTSE All World Ex US ETF	Global ex-US Equity	FLAT
50	VXUS	0.0172	Vanguard Total International Stock ETF	Global ex-US Equity	FLAT
35	QEFA	0.0171	State Street SPDR MSCI EAFE StrategicFa...	International Equity	FLAT
14	EZU	0.0170	iShares MSCI Eurozone ETF	Eurozone Equity	FLAT
22	SCZ	0.0170	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equity	FLAT
23	EFV	0.0170	iShares MSCI EAFE Value ETF	Developed ex-US Equity	FLAT
26	JPIN	0.0170	JPMorgan Diversified Return Internation...	International Equity	FLAT
38	VYMI	0.0170	Vanguard International High Dividend Yi...	International Dividend E...	FLAT
44	SCHC	0.0170	Schwab International Small-Cap Equity E...	Developed ex-US Equity	FLAT
18	IQDF	0.0169	FlexShares International Quality Divide...	International Equity	FLAT
24	FNDC	0.0169	Schwab Fundamental International Small ...	International Equity	FLAT
34	IVLU	0.0169	iShares MSCI Intl Value Factor ETF	International Equity	FLAT
37	RODM	0.0169	Hartford Multifactor Developed Markets ...	International Equity	FLAT
56	HFXI	0.0169	NYLI FTSE International Equity Currency...	International Equity	FLAT
7	DLS	0.0168	WisdomTree International SmallCap Fund	International Equity	FLAT
20	EFG	0.0168	iShares MSCI EAFE Growth ETF	Developed ex-US Equity	FLAT
36	FEZ	0.0168	State Street SPDR EURO STOXX 50 ETF	Eurozone Equity	FLAT

45	VSS	0.0168	Vanguard FTSE All-Wld ex-US SmCp Idx ETF	Global ex-US Small Cap E...	FLAT
4	FEP	0.0167	First Trust Europe AlphaDEX Fund	Europe Equity	FLAT
31	IMTM	0.0166	iShares MSCI Intl Momentum Factor ETF	International Equity	FLAT
43	VIGI	0.0166	Vanguard International Dividend Appreci...	International Dividend E...	FLAT
1	EWG	0.0164	iShares MSCI Germany Index Fund	Germany Equity	FLAT
12	EWQ	0.0164	iShares MSCI France Index Fund	France Equity	FLAT
21	GWX	0.0164	State Street SPDR S&P International Sma...	International Equity	FLAT
5	VIDI	0.0162	Vident International Equity Strategy ETF	International Equity	FLAT
59	IPKW	0.0162	Invesco International BuyBack Achievers...	International Equity	FLAT
60	EWI	0.0161	iShares MSCI Italy ETF	Italy Equity	FLAT
13	EWU	0.0160	iShares MSCI United Kingdom ETF	United Kingdom Equity	FLAT
15	IDV	0.0160	iShares International Select Dividend E...	International Dividend E...	FLAT
17	EUFN	0.0160	iShares MSCI Europe Financials ETF	Europe Financials	LONG
10	HEZU	0.0159	iShares Currency Hedged MSCI Eurozone E...	Eurozone Equity	FLAT
27	DBEF	0.0159	Xtrackers MSCI EAFE Hedged Equity ETF	Alternatives	FLAT
28	HEFA	0.0159	iShares Currency Hedged MSCI EAFE ETF	Alternatives	FLAT
42	HDEF	0.0159	Xtrackers MSCI EAFE High Dividend Yield...	International Equity	FLAT
58	EWN	0.0159	iShares MSCI Netherlands Index Fund	Netherlands Equity	LONG
55	EWD	0.0158	iShares MSCI Sweden ETF	Sweden Equity	FLAT
8	HEDJ	0.0154	WisdomTree Europe Hedged Equity Fund	Alternatives	FLAT
9	IHDG	0.0153	WisdomTree International Hedged Quality...	Alternatives	LONG
11	EWP	0.0152	iShares MSCI Spain ETF	Spain Equity	FLAT
57	EWO	0.0150	iShares MSCI Austria ETF	Austria Equity	LONG
54	EPOL	0.0137	iShares MSCI Poland ETF	Poland Equity	FLAT

Cluster 5 – U.S. Large Cap Equity

Members: 55 **Var explained:** 0.923 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +24.9%)
- Latest sell_cross: 2026-06-24
- Current 252d return: +18.5% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-24 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
U.S. Large Cap Equity	9	0.1697
U.S. Total Market Equity	7	0.1322
Global Equity	7	0.1293
Consumer Discretionary	5	0.0848
Asset Allocation	4	0.0685
U.S. Mega Cap Equity	3	0.0556
U.S. Multifactor Equity	3	0.0549
U.S. Quality / Factor Equity	3	0.0549
Communication Services	3	0.0493
U.S. ESG Equity	2	0.0376
U.S. Equity	2	0.0376
Commodity	1	0.0188
Large Cap Value Equities	1	0.0187
Large Cap Blend Equities	1	0.0186
Dividend Equity	1	0.0183
U.S. Dividend Equity	1	0.0180
U.S. Moat Equity	1	0.0172
Alternatives	1	0.0160

Replication basket – VOO (cor=1.00) basket lead at lag 1: -0.012

ticker	rep_wt	name	category	etf_signal
V00	0.96	Vanguard S&P 500 ETF	U.S. Large Cap Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
28	IWV	0.0189	iShares Russell 3000 Fund	U.S. Total Market Equity	FLAT
29	IYY	0.0189	iShares Dow Jones U.S. ETF	U.S. Total Market Equity	FLAT
34	ESGU	0.0189	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	FLAT
35	IWB	0.0189	iShares Russell 1000 ETF	U.S. Large Cap Equity	FLAT
43	VONE	0.0189	Vanguard Russell 1000 ETF	U.S. Large Cap Equity	FLAT
45	VV	0.0189	Vanguard Large-Cap ETF	U.S. Large Cap Equity	FLAT
46	ILCB	0.0189	iShares Morningstar Large-Cap ETF	U.S. Large Cap Equity	FLAT
47	ITOT	0.0189	iShares Core S&P Total U.S. Stock Marke...	U.S. Total Market Equity	FLAT
48	IVV	0.0189	iShares Core S&P 500 ETF	U.S. Large Cap Equity	FLAT
49	SCHB	0.0189	Schwab U.S. Broad Market ETF	U.S. Total Market Equity	FLAT
50	SCHX	0.0189	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	FLAT
51	SPTM	0.0189	State Street SPDR Portfolio S&P 1500 Co...	U.S. Total Market Equity	FLAT
52	VOO	0.0189	Vanguard S&P 500 ETF	U.S. Large Cap Equity	FLAT
53	VTI	0.0189	Vanguard Total Stock Market ETF	U.S. Total Market Equity	FLAT
3	SPY	0.0188	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	FLAT
32	SPYX	0.0188	State Street SPDR S&P 500 Fossil Fuel R...	U.S. Equity	FLAT
36	IWL	0.0188	iShares Russell Top 200 ETF	U.S. Equity	FLAT
40	VTHR	0.0188	Vanguard Russell 3000 ETF	U.S. Total Market Equity	FLAT
41	GSLC	0.0188	Goldman Sachs ActiveBeta U.S. Large Cap...	Commodity	FLAT
44	MGC	0.0188	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	FLAT
10	URTH	0.0187	iShares MSCI World ETF	Global Equity	FLAT
24	JHML	0.0187	John Hancock Multifactor Large Cap ETF	U.S. Large Cap Equity	FLAT
27	SUSA	0.0187	iShares ESG Optimized MSCI USA ETF	U.S. ESG Equity	FLAT
30	LRGF	0.0187	iShares U.S. Equity Factor ETF	U.S. Multifactor Equity	FLAT
42	EPS	0.0187	WisdomTree U.S. LargeCap Fund	Large Cap Value Equities	FLAT
5	ACWI	0.0186	iShares MSCI ACWI ETF	Global Equity	FLAT
13	VT	0.0186	Vanguard Total World Stock Index ETF	Global Equity	FLAT
20	TILT	0.0186	FlexShares Mornigstar US Market Factors...	U.S. Multifactor Equity	FLAT
26	DSI	0.0186	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equities	FLAT
31	OEF	0.0186	iShares S&P 100 Fund	U.S. Mega Cap Equity	FLAT
37	QUAL	0.0186	iShares MSCI USA Quality Factor ETF	U.S. Quality / Factor Eq...	FLAT
11	CRBN	0.0185	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity	FLAT
12	SPGM	0.0185	State Street SPDR Portfolio MSCI Global...	Global Equity	FLAT
6	AOA	0.0184	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation	FLAT
22	QDF	0.0183	FlexShares Quality Dividend Index Fund	Dividend Equity	FLAT
38	QUS	0.0183	State Street SPDR MSCI USA StrategicFac...	U.S. Quality / Factor Eq...	FLAT

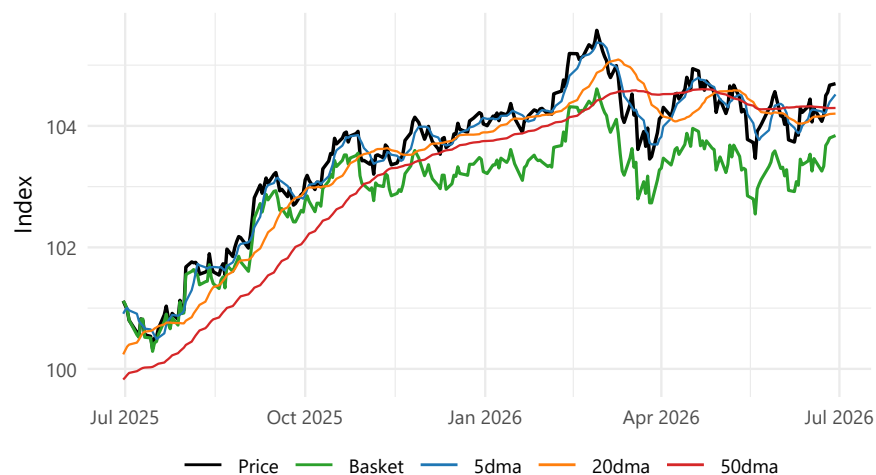
21	I00	0.0182	iShares Global 100 ETF	Global Equity	FLAT
23	CATH	0.0182	Global X S&P 500 Catholic Values ETF	Global Equity	FLAT
33	XLG	0.0182	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	FLAT
25	DGRW	0.0180	WisdomTree U.S. Quality Dividend Growth...	U.S. Dividend Equity	FLAT
39	SPHQ	0.0180	Invesco S&P 500 Quality ETF	U.S. Quality / Factor Eq...	LONG
9	AOR	0.0179	iShares Core 60/40 Balanced Allocation ...	Asset Allocation	FLAT
14	IYC	0.0175	iShares U.S. Consumer Discretionary ETF	Consumer Discretionary	FLAT
55	OMFL	0.0175	Invesco Russell 1000 Dynamic Multifacto...	U.S. Multifactor Equity	FLAT
15	VCR	0.0172	Vanguard Consumer Discretion ETF	Consumer Discretionary	FLAT
18	FDIS	0.0172	Fidelity MSCI Consumer Discretionary In...	Consumer Discretionary	FLAT
19	MOAT	0.0172	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	FLAT
1	XLY	0.0170	State Street Consumer Discretionary Sel...	Consumer Discretionary	FLAT
8	AOM	0.0168	iShares Core 40/60 Moderate Allocation ...	Asset Allocation	FLAT
16	VOX	0.0166	Vanguard Communication Services ETF	Communication Services	FLAT
17	FCOM	0.0165	Fidelity MSCI Communication Services In...	Communication Services	FLAT
2	XLC	0.0162	State Street Communication Services Sel...	Communication Services	FLAT
4	QAI	0.0160	NYLI Hedge Multi-Strategy Tracker ETF	Alternatives	FLAT
54	RTH	0.0158	VanEck Retail ETF	Consumer Discretionary	FLAT
7	AOK	0.0154	iShares Core 30/70 Conservative Allocat...	Asset Allocation	FLAT

Cluster 6 – Aggregate Fixed Income

Members: 43 **Var explained:** 0.842 **Signal:** LONG since 2026-06-25 (2 days, 0.1%)

Signal logic:

- Latest buy_cross: 2026-06-25 – filter **PASS** (252d at date = +3.8%)
- Latest sell_cross: 2026-06-08
- Current 252d return: +3.7% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-06-25 > last sell at 2026-06-08)



Category breakdown

category	n_members	abs_weight
Fixed Income	13	0.3082
Aggregate Fixed Income	6	0.1478
Investment Grade Corporate Bond	6	0.1441
Mortgage-Backed Fixed Income	6	0.1326
Active Fixed Income	4	0.0943
Cash / Ultra-Short Fixed Income	3	0.0667
Target Maturity Bond	2	0.0420
International Fixed Income	2	0.0417
Agency Fixed Income	1	0.0227

Replication basket – BND (cor=0.99) basket lead at lag 1: 0.009

ticker	rep_wt	name	category	etf_signal
BND	0.729	Vanguard Total Bond Market ETF	Aggregate Fixed Income	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	BIV	0.0250	Vanguard Intermediate-Term Bond ETF	Fixed Income	LONG
16	AGG	0.0249	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	LONG
17	BND	0.0249	Vanguard Total Bond Market ETF	Aggregate Fixed Income	LONG
33	VGIT	0.0249	Vanguard Intermediate-Term Treasury ETF	Fixed Income	LONG
19	SCHZ	0.0248	Schwab US Aggregate Bond ETF	Aggregate Fixed Income	LONG
29	IEI	0.0248	iShares 3-7 Year Treasury Bond ETF	Fixed Income	LONG
30	SPTI	0.0248	State Street SPDR Portfolio Intermediat...	Fixed Income	LONG
32	SCHR	0.0248	Schwab Intermediate-Term U.S. Treasury ...	Fixed Income	LONG
6	GVI	0.0247	iShares Intermediate Government/Credit ...	Fixed Income	LONG
12	IUSB	0.0247	iShares Core Universal USD Bond ETF	Aggregate Fixed Income	LONG
20	SPAB	0.0247	State Street SPDR Portfolio Aggregate B...	Aggregate Fixed Income	LONG
1	IEF	0.0246	iShares 7-10 Year Treasury Bond ETF	Fixed Income	LONG
11	ISTB	0.0245	iShares Core 1-5 Year USD Bond ETF	Fixed Income	LONG
9	SPIB	0.0243	State Street SPDR Portfolio Intermediat...	Investment Grade Corpora...	LONG
10	IGIB	0.0243	iShares 5-10 Year Investment Grade Corp...	Investment Grade Corpora...	LONG
14	BSV	0.0242	Vanguard Short-Term Bond ETF	Fixed Income	FLAT
15	VCIT	0.0242	Vanguard Intermediate-Term Corporate Bo...	Investment Grade Corpora...	LONG
35	MBB	0.0242	iShares MBS ETF	Mortgage-Backed Fixed In...	LONG
36	VMBS	0.0242	Vanguard Mortgage-Backed Securities ETF	Mortgage-Backed Fixed In...	LONG
5	FBND	0.0241	Fidelity Total Bond ETF	Active Fixed Income	LONG
37	SPMB	0.0241	State Street SPDR Portfolio Mortgage Ba...	Mortgage-Backed Fixed In...	LONG
24	IGSB	0.0240	iShares 1-5 Year Investment Grade Corpo...	Investment Grade Corpora...	LONG
26	VCSH	0.0239	Vanguard Short-Term Corporate Bond ETF	Investment Grade Corpora...	LONG
3	BOND	0.0238	PIMCO Active Bond Exchange-Traded Fund ...	Active Fixed Income	LONG
4	GTO	0.0238	Invesco Total Return Bond ETF	Active Fixed Income	LONG
7	AGGY	0.0238	WisdomTree Yield Enhanced U.S. Aggregat...	Aggregate Fixed Income	LONG
25	SLQD	0.0234	iShares 0-5 Year Investment Grade Corpo...	Investment Grade Corpora...	LONG
31	GOVT	0.0233	iShares U.S. Treasury Bond ETF	Fixed Income	LONG
34	GNMA	0.0233	iShares GNMA Bond ETF	Mortgage-Backed Fixed In...	LONG
38	SHY	0.0229	iShares 1-3 Year Treasury Bond ETF	Fixed Income	FLAT
28	AGZ	0.0227	iShares Agency Bond ETF	Agency Fixed Income	LONG
18	TOTL	0.0226	State Street DoubleLine Total Return Ta...	Active Fixed Income	LONG
23	SPSB	0.0226	State Street SPDR Portfolio Short Term ...	Fixed Income	LONG
41	VGSH	0.0225	Vanguard Short-Term Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
39	SPTS	0.0221	State Street SPDR Portfolio Short Term ...	Cash / Ultra-Short Fixed...	FLAT
40	SCHO	0.0221	Schwab Short-Term U.S. Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
22	BSCQ	0.0213	Invesco BulletShares 2026 Corporate Bon...	Target Maturity Bond	FLAT
42	BNDX	0.0210	Vanguard Total International Bond ETF	International Fixed Inco...	LONG
8	IBDR	0.0207	iShares iBonds Dec 2026 Term Corporate ...	Target Maturity Bond	FLAT
43	IAGG	0.0207	iShares International Aggregate Bond Fu...	International Fixed Inco...	LONG
27	CMBS	0.0186	iShares CMBS Bond ETF	Mortgage-Backed Fixed In...	LONG
2	LMBS	0.0181	First Trust Low Duration Opportunities ...	Mortgage-Backed Fixed In...	LONG
21	LDUR	0.0172	PIMCO Enhanced Low Duration Active Exch...	Fixed Income	LONG

Cluster 7 – Emerging Markets Equity**Members:** 25 **Var explained:** 0.873 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +57.0%)
- Latest sell_cross: 2026-06-29
- Current 252d return: +33.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-29 more recent than last effective buy 2026-04-10)

**Category breakdown**

category	n_members	abs_weight
Emerging Markets Equity	19	0.7727
Asia Equity	2	0.0814
Asia ex-Japan Equity	1	0.0418
Taiwan Equity	1	0.0361
South Korea Equity	1	0.0341
Global Dividend Equity	1	0.0340

Replication basket – IEMG (cor=0.99) basket lead at lag 1: -0.029

ticker	rep_wt	name	category	etf_signal
IEMG	0.922	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equi...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	IEMG	0.0425	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equity	FLAT
2	EEM	0.0424	iShares MSCI Emerging Index Fund	Emerging Markets Equity	FLAT
23	SPEM	0.0423	State Street SPDR Portfolio Emerging Ma...	Emerging Markets Equity	FLAT
3	VWO	0.0422	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equity	FLAT
16	GEM	0.0422	Goldman Sachs ActiveBeta Emerging Marke...	Emerging Markets Equity	FLAT
20	ESGE	0.0422	iShares ESG Aware MSCI EM ETF	Emerging Markets Equity	FLAT
22	SCHE	0.0421	Schwab Emerging Markets Equity ETF	Emerging Markets Equity	FLAT
18	XSOE	0.0420	WisdomTree Emerging Markets Ex-State Ow...	Emerging Markets Equity	FLAT
5	AAXJ	0.0418	iShares MSCI All Country Asia ex Japan ...	Asia ex-Japan Equity	FLAT
15	EMGF	0.0418	iShares Emerging Markets Equity Factor ...	Emerging Markets Equity	FLAT
11	EEMA	0.0414	iShares MSCI Emerging Markets Asia ETF	Emerging Markets Equity	FLAT
14	GMF	0.0411	State Street SPDR S&P Emerging Asia Pac...	Asia Equity	FLAT
17	FNDE	0.0410	Schwab Fundamental Emerging Markets Equ...	Emerging Markets Equity	FLAT
12	PXH	0.0409	Invesco RAFI Emerging Markets ETF	Emerging Markets Equity	FLAT
19	EEMV	0.0405	iShares MSCI Emerging Markets Min Vol F...	Emerging Markets Equity	FLAT
8	DGS	0.0402	WisdomTree Emerging Market SmallCap Fund	Emerging Markets Equity	FLAT
10	AIA	0.0402	iShares Asia 50 ETF	Asia Equity	FLAT
7	DEM	0.0396	WisdomTree Emerging Markets High Divide...	Emerging Markets Equity	FLAT
6	EWX	0.0392	State Street SPDR S&P Emerging Markets ...	Emerging Markets Equity	LONG
4	FEM	0.0388	First Trust Emerging Markets AlphaDEX F...	Emerging Markets Equity	FLAT
13	DVYE	0.0365	iShares Emerging Markets Dividend Index...	Emerging Markets Equity	FLAT
9	EWT	0.0361	iShares MSCI Taiwan ETF	Taiwan Equity	LONG
25	FEMS	0.0348	First Trust Emerging Markets Small Cap ...	Emerging Markets Equity	FLAT
1	EWY	0.0341	iShares MSCI South Korea ETF	South Korea Equity	FLAT
24	SDIV	0.0340	Global X SuperDividend ETF	Global Dividend Equity	FLAT

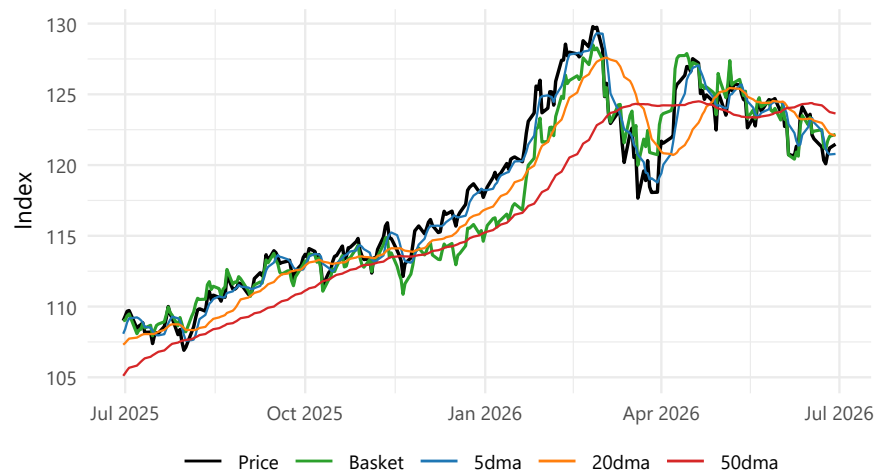
Cluster 8 – Asia-Pacific Equity

Members: 24 **Var explained:** 0.683 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +32.3%)

- Latest sell_cross: 2026-06-22
- Current 252d return: +12.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-22 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Asia-Pacific Equity	2	0.0937
International Dividend Equity	2	0.0921
International Equity	2	0.0920
Global Equity	2	0.0883
Real Estate	2	0.0882
Emerging Markets Fixed Income	2	0.0803
Currency	2	0.0776
Australia Equity	1	0.0465
Canada Equity	1	0.0442
Switzerland Equity	1	0.0416
Belgium Equity	1	0.0414
South Africa Equity	1	0.0393
Latin America Equity	1	0.0378
New Zealand Equity	1	0.0377
Mexico Equity	1	0.0358
Brazil Equity	1	0.0319
Chile Equity	1	0.0318

Replication basket – EWZ+IGF+BBAX (cor=0.97) basket lead at lag 1: -0.002

ticker	rep_wt	name	category	etf_signal
EWZ	0.091	iShares MSCI Brazil ETF	Brazil Equity	FLAT
IGF	0.298	iShares Global Infrastructure ETF	Global Equity	LONG
BBAX	0.451	JPMorgan BetaBuilders Developed Asia ...	Asia-Pacific Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

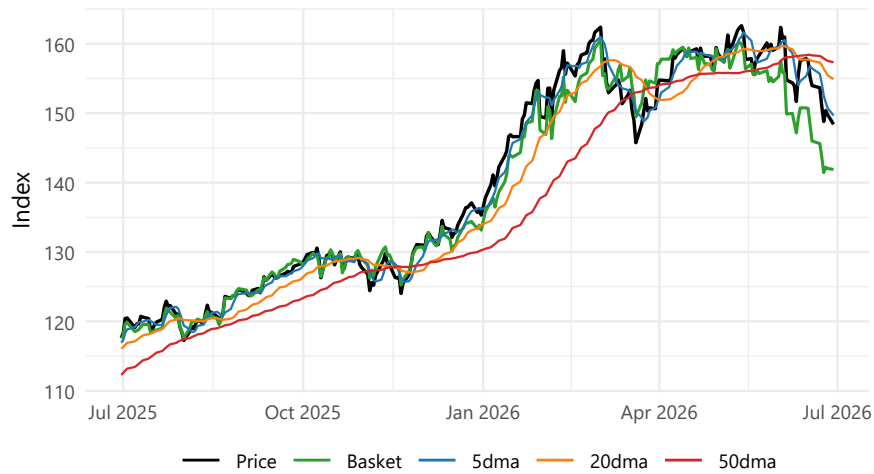
	ticker	PC_wt	name	category	etf_signal
11	IDLV	0.0469	Invesco S&P International Developed Low...	International Equity	FLAT
7	EPP	0.0468	iShares MSCI Pacific Ex-Japan Index Fund	Asia-Pacific Equity	FLAT
24	BBAX	0.0468	JPMorgan BetaBuilders Developed Asia Pa...	Asia-Pacific Equity	FLAT
4	EWA	0.0465	iShares MSCI Australia Index Fund	Australia Equity	FLAT
10	DWX	0.0464	State Street SPDR S&P International Div...	International Dividend E...	FLAT
3	PID	0.0457	Invesco International Dividend Achiever...	International Dividend E...	FLAT
12	EFAV	0.0451	iShares MSCI EAFE Min Vol Factor ETF	International Equity	FLAT
8	NFRA	0.0448	FlexShares STOXX Global Broad Infrastru...	Global Equity	FLAT
13	VNQI	0.0446	Vanguard Global ex-U.S. Real Estate ETF	Real Estate	FLAT
6	EWK	0.0442	iShares MSCI Canada Index Fund	Canada Equity	FLAT
2	RWX	0.0435	State Street SPDR Dow Jones Internation...	Real Estate	FLAT
9	IGF	0.0435	iShares Global Infrastructure ETF	Global Equity	LONG
5	EWL	0.0416	iShares MSCI Switzerland ETF	Switzerland Equity	LONG
23	EWK	0.0414	iShares MSCI Belgium ETF	Belgium Equity	FLAT
19	FXA	0.0411	Invesco CurrencyShares Australian Dolla...	Currency	FLAT
14	EBND	0.0402	State Street SPDR Bloomberg Emerging Ma...	Emerging Markets Fixed I...	LONG
15	EMLC	0.0401	VanEck J. P. Morgan EM Local Currency B...	Emerging Markets Fixed I...	LONG
21	EZA	0.0393	iShares MSCI South Africa Index Fund	South Africa Equity	FLAT
18	ILF	0.0378	iShares Latin America 40 ETF	Latin America Equity	FLAT
20	ENZL	0.0377	iShares MSCI New Zealand ETF	New Zealand Equity	FLAT
22	FXC	0.0364	Invesco CurrencyShares Canadian Dollar ...	Currency	FLAT
16	EWV	0.0358	iShares MSCI Mexico ETF	Mexico Equity	FLAT
1	EWZ	0.0319	iShares MSCI Brazil ETF	Brazil Equity	FLAT
17	ECH	0.0318	iShares MSCI Chile ETF	Chile Equity	FLAT

Cluster 9 – Materials

Members: 21 **Var explained:** 0.702 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +44.4%)
- Latest sell_cross: 2026-06-09
- Current 252d return: +26.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-09 more recent than last effective buy 2026-04-07)



Category breakdown

category	n_members	abs_weight
Materials	13	0.6545
MLP / Energy Infrastructure	4	0.1687
Energy	1	0.0502
Consumer Staples	1	0.0493
Norway Equity	1	0.0438
Industrial Metals	1	0.0336

Replication basket – AMLP+URA+GUNR (cor=0.96) basket lead at lag 1: 0.043

ticker	rep_wt	name	category	etf_signal
AMLP	0.109	Alerian MLP ETF	MLP / Energy Infrastr...	FLAT
URA	0.082	Global X Uranium ETF	Materials	FLAT
GUNR	0.776	FlexShares Global Upstream Natural Re...	Materials	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

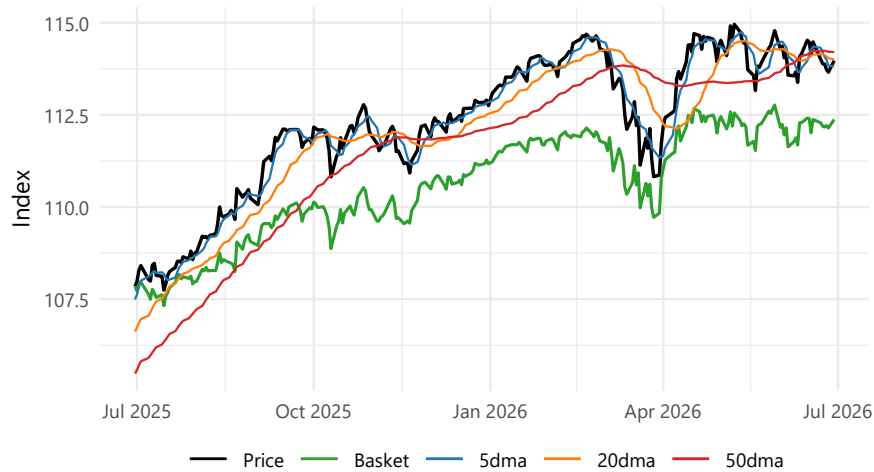
	ticker	PC_wt	name	category	etf_signal
15	GNR	0.0550	State Street SPDR S&P Global Natural Re...	Materials	FLAT
14	GUNR	0.0543	FlexShares Global Upstream Natural Reso...	Materials	FLAT
10	MXI	0.0529	iShares Global Materials ETF	Materials	FLAT
11	IYM	0.0529	iShares U.S. Basic Materials ETF	Materials	FLAT
21	NANR	0.0525	State Street SPDR S&P North American Na...	Materials	FLAT
13	VAW	0.0514	Vanguard Materials ETF	Materials	LONG
16	PICK	0.0509	iShares MSCI Global Select Metals & Min...	Materials	FLAT
12	RSPM	0.0506	Invesco S&P 500 Equal Weight Materials ...	Materials	LONG
1	XLB	0.0505	State Street Materials Select Sector SP...	Materials	LONG
20	IGE	0.0502	iShares North American Natural Resource...	Energy	FLAT
4	XME	0.0499	State Street SPDR S&P Metals & Mining E...	Materials	FLAT
17	SLX	0.0498	VanEck Steel ETF	Materials	FLAT
6	MOO	0.0493	VanEck Agribusiness ETF	Consumer Staples	FLAT
3	COPX	0.0472	Global X Copper Miners ETF	Materials	FLAT
18	NORW	0.0438	Global X MSCI Norway ETF	Norway Equity	FLAT
5	EMLP	0.0437	First Trust North American Energy Infra...	MLP / Energy Infrastruct...	LONG
9	MLPX	0.0436	Global X MLP & Energy Infrastructure ETF	MLP / Energy Infrastruct...	LONG
7	AMLP	0.0413	Alerian MLP ETF	MLP / Energy Infrastruct...	FLAT
8	MLPA	0.0401	Global X MLP ETF	MLP / Energy Infrastruct...	LONG
2	URA	0.0367	Global X Uranium ETF	Materials	FLAT
19	DBB	0.0336	Invesco DB Base Metals Fund	Industrial Metals	FLAT

Cluster 10 – High Yield Bond

Members: 21 **Var explained:** 0.766 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-06-17 – filter **PASS** (252d at date = +7.3%)
- Latest sell_cross: 2026-06-26
- Current 252d return: +5.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-26 more recent than last effective buy 2026-06-17)



Category breakdown

category	n_members	abs_weight
High Yield Bond	10	0.5156
Preferred Stock	7	0.3060
Emerging Markets Fixed Income	4	0.1784

Replication basket – HYG (cor=0.95) basket lead at lag 1: 0.047

ticker	rep_wt	name	category	etf_signal
HYG	0.917	iShares iBoxx \$ High Yield Corporate ...	High Yield Bond	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	HYG	0.0526	iShares iBoxx \$ High Yield Corporate Bo...	High Yield Bond	LONG
13	JNK	0.0526	State Street SPDR Bloomberg High Yield ...	High Yield Bond	LONG
19	HYLB	0.0524	Xtrackers USD High Yield Corporate Bond...	High Yield Bond	LONG
18	BBHY	0.0523	JPMorgan BetaBuilders USD High Yield Co...	High Yield Bond	LONG
16	SHYG	0.0521	iShares 0-5 Year High Yield Corporate B...	High Yield Bond	LONG

14	SJNK	0.0520	State Street SPDR Bloomberg Short Term ...	High Yield Bond	LONG
15	ANGL	0.0516	VanEck Fallen Angel High Yield Bond ETF	High Yield Bond	LONG
17	FALN	0.0516	iShares Fallen Angels USD Bond ETF	High Yield Bond	LONG
6	HYS	0.0511	PIMCO 0-5 Year High Yield Corporat Bond...	High Yield Bond	LONG
9	PFF	0.0479	iShares Preferred and Income Securities...	Preferred Stock	FLAT
3	HYLS	0.0473	First Trust Tactical High Yield ETF	High Yield Bond	FLAT
2	EMB	0.0468	iShares J.P. Morgan USD Emerging Market...	Emerging Markets Fixed I...	FLAT
21	VWOB	0.0467	Vanguard Emerging Markets Government Bo...	Emerging Markets Fixed I...	LONG
11	PFXF	0.0461	VanEck Preferred Securities ex Financia...	Preferred Stock	FLAT
20	PCY	0.0458	Invesco Emerging Markets Sovereign Debt...	Emerging Markets Fixed I...	LONG
7	PGX	0.0446	Invesco Preferred ETF	Preferred Stock	FLAT
10	PSK	0.0441	State Street SPDR ICE Preferred Securit...	Preferred Stock	FLAT
4	FPE	0.0433	First Trust Preferred Securities and In...	Preferred Stock	FLAT
5	PGF	0.0431	Invesco Financial Preferred ETF	Preferred Stock	FLAT
12	HYEM	0.0392	VanEck Emerging Markets High Yield Bond...	Emerging Markets Fixed I...	LONG
8	VRP	0.0368	Invesco Variable Rate Preferred ETF	Preferred Stock	FLAT

Cluster 11 – Healthcare

Members: 17 **Var explained:** 0.761 **Signal:** LONG since 2026-05-06 (36 days, 11.1%)

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +13.8%)
- Latest sell_cross: 2026-04-29
- Current 252d return: +26.6% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-06 > last sell at 2026-04-29)



Category breakdown

category	n_members	abs_weight
Healthcare	16	0.9418
U.S. Small Cap Healthcare	1	0.0582

Replication basket – VHT (cor=0.93) basket lead at lag 1: 0.000

ticker	rep_wt	name	category	etf_signal
VHT	1.097	Vanguard Health Care ETF	Healthcare	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
15	VHT	0.0654	Vanguard Health Care ETF	Healthcare	LONG
16	FHLC	0.0653	Fidelity MSCI Health Care Index ETF	Healthcare	LONG
9	FXH	0.0646	First Trust Health Care AlphaDEX	Healthcare	LONG
14	RSPH	0.0640	Invesco S&P 500 Equal Weight Health Car...	Healthcare	LONG
13	IYH	0.0638	iShares U.S. Healthcare ETF	Healthcare	LONG
10	IXJ	0.0623	iShares Global Healthcare ETF	Healthcare	FLAT

1	XLV	0.0621	State Street Health Care Select Sector ...	Healthcare	LONG
3	IBB	0.0604	iShares Biotechnology ETF	Healthcare	LONG
6	FBT	0.0596	First Trust Amex Biotech Index Fund	Healthcare	FLAT
8	PSCH	0.0582	Invesco S&P SmallCap Health Care ETF	U.S. Small Cap Healthcare	LONG
7	XHE	0.0574	State Street SPDR S&P Health Care Equip...	Healthcare	LONG
12	IHI	0.0570	iShares U.S. Medical Devices ETF	Healthcare	FLAT
17	PPH	0.0544	VanEck Pharmaceutical ETF	Healthcare	LONG
5	PTH	0.0542	Invesco Dorsey Wright Healthcare Moment...	Healthcare	LONG
4	XBI	0.0534	State Street SPDR S&P Biotech ETF	Healthcare	LONG
2	ARKG	0.0508	ARK Genomic Revolution ETF	Healthcare	FLAT
11	IHF	0.0472	iShares U.S. Health Care Providers ETF	Healthcare	LONG

Cluster 12 – China Equity

Members: 15 **Var explained:** 0.800 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-15 – filter **PASS** (252d at date = +24.3%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +3.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-04-15)



Category breakdown

category	n_members	abs_weight
China Equity	13	0.8766
Emerging Markets Equity	1	0.0643
Hong Kong Equity	1	0.0591

Replication basket – KWEB (cor=0.92) basket lead at lag 1: 0.012

ticker	rep_wt	name	category	etf_signal
KWEB	0.563	KraneShares CSI China Internet ETF	China Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

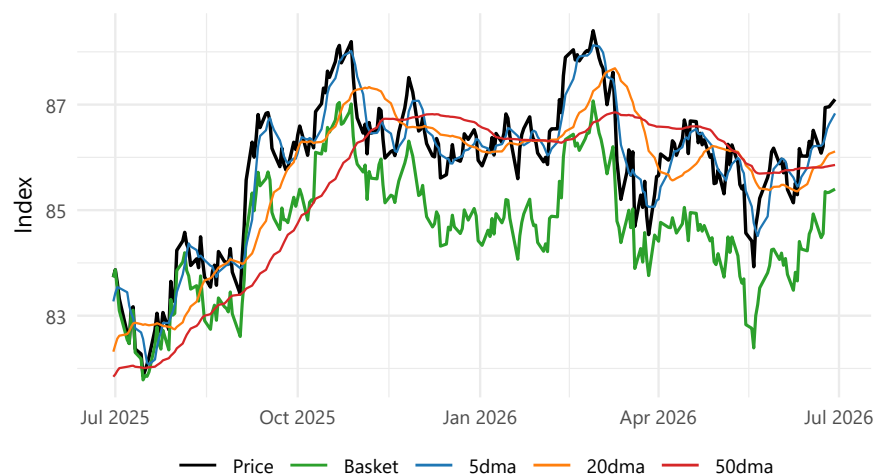
	ticker	PC_wt	name	category	etf_signal
11	CXSE	0.0738	WisdomTree China ex-State-Owned Enterpr...	China Equity	FLAT
9	GXC	0.0736	State Street SPDR S&P China ETF	China Equity	FLAT
1	MCHI	0.0730	iShares MSCI China ETF	China Equity	FLAT
2	FXI	0.0712	iShares China Large-Cap ETF	China Equity	FLAT
5	CQQQ	0.0712	Invesco China Technology ETF	China Equity	FLAT
7	CHIQ	0.0705	Global X MSCI China Consumer Discretion...	China Equity	FLAT
4	KWEB	0.0680	KraneShares CSI China Internet ETF	China Equity	FLAT
6	ASHR	0.0668	Xtrackers Harvest CSI 300 China A-Share...	China Equity	LONG
8	CNYA	0.0661	iShares MSCI China A ETF	China Equity	LONG
13	KGRN	0.0652	KraneShares MSCI China Clean Technology...	China Equity	FLAT
3	EMQQ	0.0643	EMQQ The Emerging Markets Internet ETF	Emerging Markets Equity	FLAT
10	KBA	0.0596	KraneShares Boseri MSCI China A 50 Conn...	China Equity	FLAT
14	CNXT	0.0593	VanEck ChiNext Innovators ETF	China Equity	FLAT
12	EWK	0.0591	iShares MSCI Hong Kong Index Fund	Hong Kong Equity	FLAT
15	KURE	0.0584	KraneShares MSCI All China Health Care ...	China Equity	FLAT

Cluster 13 – Fixed Income

Members: 15 **Var explained:** 0.928 **Signal:** LONG since 2026-06-15 (9 days, 1.3%)

Signal logic:

- Latest buy_cross: 2026-06-15 – filter **PASS** (252d at date = +3.9%)
- Latest sell_cross: 2026-05-13
- Current 252d return: +4.4% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-06-15 > last sell at 2026-05-13)



Category breakdown

category	n_members	abs_weight
Investment Grade Corporate Bond	7	0.4706
Fixed Income	6	0.4009
Active Fixed Income	1	0.0665
Taxable Municipal Bond	1	0.0619

Replication basket – TLT (cor=0.97) basket lead at lag 1: -0.042

ticker	rep_wt	name	category	etf_signal
TLT	0.694	iShares 20+ Year Treasury Bond ETF	Fixed Income	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

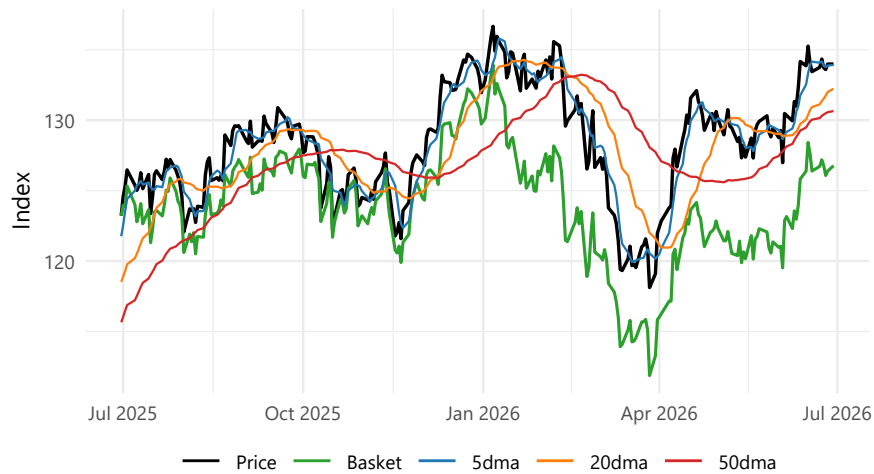
	ticker	PC_wt	name	category	etf_signal
9	BLV	0.0687	Vanguard Long-Term Bond ETF	Fixed Income	LONG
7	SPLB	0.0678	State Street SPDR Portfolio Long Term C...	Investment Grade Corpora...	LONG
8	IGLB	0.0678	iShares 10 Year Investment Grade Corpor...	Investment Grade Corpora...	LONG
10	VCLT	0.0676	Vanguard Long-Term Corporate Bond ETF	Investment Grade Corpora...	LONG
2	LQD	0.0675	iShares iBoxx \$ Investment Grade Corpor...	Investment Grade Corpora...	LONG
6	QLTA	0.0671	iShares Aaa A Rated Corporate Bond ETF	Investment Grade Corpora...	LONG
11	USIG	0.0671	iShares Broad USD Investment Grade Corp...	Investment Grade Corpora...	LONG
12	TLH	0.0668	iShares 10-20 Year Treasury Bond ETF	Fixed Income	LONG
14	SPTL	0.0667	State Street SPDR Portfolio Long Term T...	Fixed Income	LONG
15	VGLT	0.0666	Vanguard Long-Term Treasury ETF	Fixed Income	LONG
3	FLCO	0.0665	Franklin Investment Grade Corporate ETF	Active Fixed Income	LONG
1	TLT	0.0664	iShares 20+ Year Treasury Bond ETF	Fixed Income	LONG
5	CORP	0.0657	PIMCO Investment Grade Corporate Bond I...	Investment Grade Corpora...	LONG
13	EDV	0.0656	Vanguard Extended Duration Treasury ETF	Fixed Income	LONG
4	BAB	0.0619	Invesco Taxable Municipal Bond ETF	Taxable Municipal Bond	LONG

Cluster 14 – Financials

Members: 14 **Var explained:** 0.875 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +19.9%)
- Latest sell_cross: 2026-05-11
- Current 252d return: +9.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-11 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Financials	13	0.9382
Global Equity	1	0.0618

Replication basket – XLF (cor=0.96) basket lead at lag 1: 0.011

ticker	rep_wt	name	category	etf_signal
XLF	1.096	State Street Financial Select Sector ...	Financials	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
10	VFH	0.0755	Vanguard Financials ETF	Financials	FLAT
11	FNCL	0.0755	Fidelity MSCI Financials Index ETF	Financials	LONG
5	FXO	0.0752	First Trust Financials AlphaDEX	Financials	LONG
7	IYF	0.0752	iShares U.S. Financial ETF	Financials	FLAT
8	IYG	0.0748	iShares U.S. Financial Services ETF	Financials	FLAT
1	XLF	0.0741	State Street Financial Select Sector SP...	Financials	LONG

9	KBWB	0.0733	Invesco KBW Bank ETF	Financials	FLAT
3	KBE	0.0712	State Street SPDR S&P Bank ETF	Financials	FLAT
14	KCE	0.0707	State Street SPDR S&P Capital Markets E...	Financials	FLAT
6	IAT	0.0706	iShares U.S. Regional Banks ETF	Financials	LONG
13	IAI	0.0698	iShares U.S. Broker-Dealers & Securitie...	Financials	FLAT
2	KRE	0.0687	State Street SPDR S&P Regional Banking ...	Financials	FLAT
4	KIE	0.0636	State Street SPDR S&P Insurance ETF	Financials	LONG
12	PSP	0.0618	Invesco Global Listed Private Equity ETF	Global Equity	FLAT

Cluster 15 – Real Estate

Members: 14 **Var explained:** 0.961 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +21.2%)
- Latest sell_cross: 2026-06-03
- Current 252d return: +15.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-03 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Real Estate	14	1

Replication basket – VNQ (cor=0.99) basket lead at lag 1: -0.000

ticker	rep_wt	name	category	etf_signal
VNQ	0.97	Vanguard Real Estate ETF	Real Estate	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

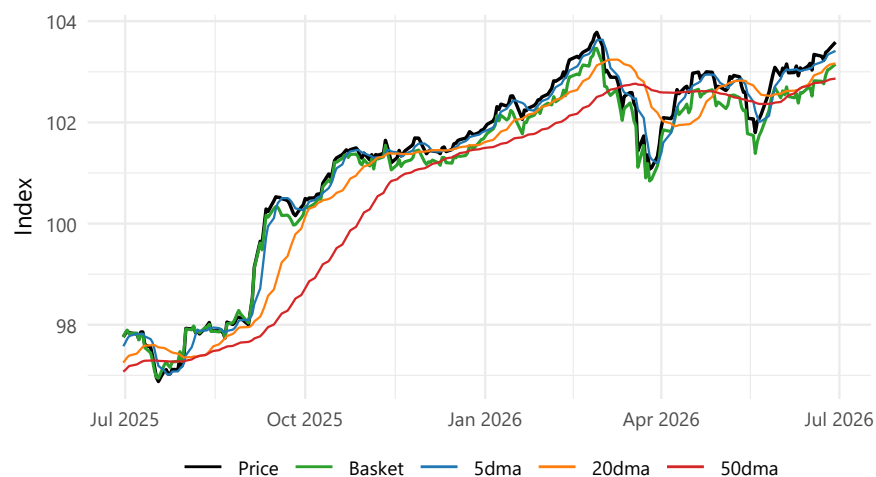
	ticker	PC_wt	name	category	etf_signal
8	VNQ	0.0725	Vanguard Real Estate ETF	Real Estate	FLAT
9	FREL	0.0725	Fidelity MSCI Real Estate Index ETF	Real Estate	LONG
11	SCHH	0.0725	Schwab U.S. REIT ETF	Real Estate	FLAT
2	IYR	0.0724	iShares U.S. Real Estate ETF	Real Estate	LONG
10	USRT	0.0724	iShares Core U.S. REIT ETF	Real Estate	FLAT
6	RWR	0.0723	State Street SPDR Dow Jones REIT ETF	Real Estate	FLAT
13	PSR	0.0722	Invesco Active U.S. Real Estate Fund	Real Estate	FLAT
1	XLRE	0.0719	State Street Real Estate Select Sector ...	Real Estate	LONG
5	ICF	0.0719	iShares Select U.S. REIT ETF	Real Estate	FLAT
7	REET	0.0715	iShares Global REIT ETF	Real Estate	FLAT
3	RWO	0.0711	State Street SPDR Dow Jones Global Real...	Real Estate	LONG
4	REZ	0.0698	iShares Residential and Multisector Rea...	Real Estate	FLAT
14	NURE	0.0688	Nuveen Short-Term REIT ETF	Real Estate	LONG
12	INDS	0.0682	Pacer Industrial Real Estate ETF	Real Estate	LONG

Cluster 16 – Municipal Bond

Members: 13 **Var explained:** 0.819 **Signal:** LONG since 2026-05-28 (21 days, 0.6%)

Signal logic:

- Latest buy_cross: 2026-05-28 – filter **PASS** (252d at date = +5.9%)
- Latest sell_cross: 2026-05-15
- Current 252d return: +6.2% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-28 > last sell at 2026-05-15)



Category breakdown

category	n_members	abs_weight
Municipal Bond	13	1

Replication basket – MUB (cor=0.95) basket lead at lag 1: 0.178

ticker	rep_wt	name	category	etf_signal
MUB	0.904	iShares National Muni Bond ETF	Municipal Bond	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	VTEB	0.0817	Vanguard Tax-Exempt Bond ETF	Municipal Bond	LONG
11	MUB	0.0813	iShares National Muni Bond ETF	Municipal Bond	LONG
6	CMF	0.0802	iShares California Muni Bond ETF	Municipal Bond	LONG
9	TFI	0.0802	State Street SPDR Nuveen ICE Municipal ...	Municipal Bond	LONG
8	ITM	0.0800	VanEck Intermediate Muni ETF	Municipal Bond	LONG
1	FMB	0.0795	First Trust Managed Municipal ETF	Municipal Bond	LONG
7	NYF	0.0793	iShares New York Muni Bond ETF	Municipal Bond	LONG

5	PZA	0.0780	Invesco National AMT-Free Municipal Bon...	Municipal Bond	LONG
4	MUNI	0.0769	PIMCO Intermediate Municipal Bond Activ...	Municipal Bond	LONG
2	HYD	0.0748	VanEck High Yield Muni ETF	Municipal Bond	LONG
3	HYMB	0.0720	State Street SPDR Nuveen ICE High Yield...	Municipal Bond	LONG
12	SUB	0.0695	iShares Short-Term National Muni Bond E...	Municipal Bond	LONG
10	SHM	0.0667	State Street SPDR Nuveen ICE Short Term...	Municipal Bond	LONG

Cluster 17 – Gold

Members: 12 **Var explained:** 0.861 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-12 – filter **PASS** (252d at date = +75.1%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +27.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-05-12)



Category breakdown

category	n_members	abs_weight
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Materials	5	0.4157
Gold	3	0.2502
Precious Metals	2	0.1724
Silver	2	0.1617

Replication basket – GLD+IAU+SLV (cor=0.95) basket lead at lag 1: -0.010

ticker	rep_wt	name	category	etf_signal
GLD	0.022	SPDR Gold Shares	Gold	FLAT
IAU	0.750	iShares Gold Trust Shares	Gold	FLAT
SLV	0.411	iShares Silver Trust	Silver	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	DBP	0.0866	Invesco DB Precious Metals Fund	Precious Metals	FLAT
7	GLTR	0.0859	abrdn Physical Precious Metals Basket S...	Precious Metals	FLAT
2	GDXJ	0.0841	VanEck Junior Gold Miners ETF	Materials	FLAT
1	GDX	0.0840	VanEck Gold Miners ETF	Materials	FLAT
3	GLD	0.0834	SPDR Gold Shares	Gold	FLAT
9	IAU	0.0834	iShares Gold Trust Shares	Gold	FLAT
10	SGOL	0.0833	abrdn Physical Gold Shares ETF	Gold	FLAT
12	GOAU	0.0831	US Global GO Gold and Precious Metal Mi...	Materials	FLAT
6	SIL	0.0828	Global X Silver Miners ETF	Materials	FLAT
5	SILJ	0.0818	Amplify Junior Silver Miners ETF	Materials	FLAT
4	SLV	0.0809	iShares Silver Trust	Silver	LONG
8	SIVR	0.0808	abrdn Physical Silver Shares ETF	Silver	LONG

Cluster 18 – Consumer Staples

Members: 11 **Var explained:** 0.744 **Signal:** LONG since 2026-06-26 (1 days, -0.4%)

Signal logic:

- Latest buy_cross: 2026-06-26 – filter **PASS** (252d at date = +9.9%)
- Latest sell_cross: 2026-05-29
- Current 252d return: +9.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-06-26 > last sell at 2026-05-29)



Category breakdown

category	n_members	abs_weight
Consumer Staples	6	0.5733
Utilities	4	0.3519
U.S. Small Cap Consumer Staples	1	0.0748

Replication basket – XLU+XLP (cor=0.99) basket lead at lag 1: -0.011

ticker	rep_wt	name	category	etf_signal
XLU	0.364	State Street Utilities Select Sector ...	Utilities	LONG
XLP	0.588	State Street Consumer Staples Select ...	Consumer Staples	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	VDC	0.0983	Vanguard Consumer Staples ETF	Consumer Staples	LONG
6	FSTA	0.0983	Fidelity MSCI Consumer Staples Index ETF	Consumer Staples	LONG
1	XLP	0.0975	State Street Consumer Staples Select Se...	Consumer Staples	LONG
4	IYK	0.0949	iShares U.S. Consumer Staples ETF	Consumer Staples	LONG

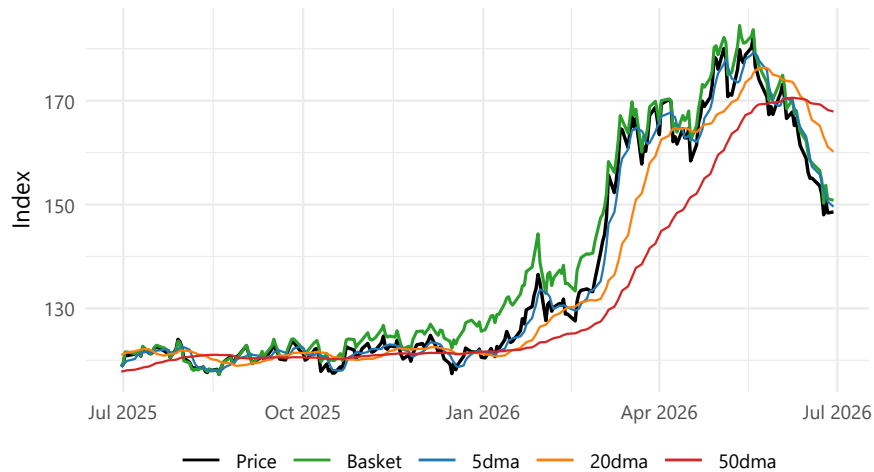
3	KXI	0.0946	iShares Global Consumer Staples ETF	Consumer Staples	LONG
11	PBJ	0.0897	Invesco Food & Beverage ETF	Consumer Staples	FLAT
7	IDU	0.0890	iShares U.S. Utilities ETF	Utilities	LONG
9	FUTY	0.0878	Fidelity MSCI Utilities Index ETF	Utilities	LONG
8	VPU	0.0877	Vanguard Utilities ETF	Utilities	LONG
2	XLU	0.0874	State Street Utilities Select Sector SP...	Utilities	LONG
10	PSCC	0.0748	Invesco S&P SmallCap Consumer Staples E...	U.S. Small Cap Consumer ...	LONG

Cluster 19 – Broad Commodities

Members: 11 **Var explained:** 0.772 **Signal:** LONG since 2026-06-03 (17 days, -14.2%)

Signal logic:

- Latest buy_cross: 2026-06-03 – filter **PASS** (252d at date = +48.1%)
- Latest sell_cross: 2026-05-27
- Current 252d return: +24.5% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-06-03 > last sell at 2026-05-27)



Category breakdown

category	n_members	abs_weight
Broad Commodities	6	0.6172
Oil	2	0.1980
Energy Commodities	1	0.1019
Agricultural Commodities	1	0.0499
Natural Gas	1	0.0329

Replication basket – PDBC (cor=0.98) basket lead at lag 1: 0.075

ticker	rep_wt	name	category	etf_signal
PDBC	1.121	Invesco Optimum Yield Diversified Com...	Broad Commodities	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

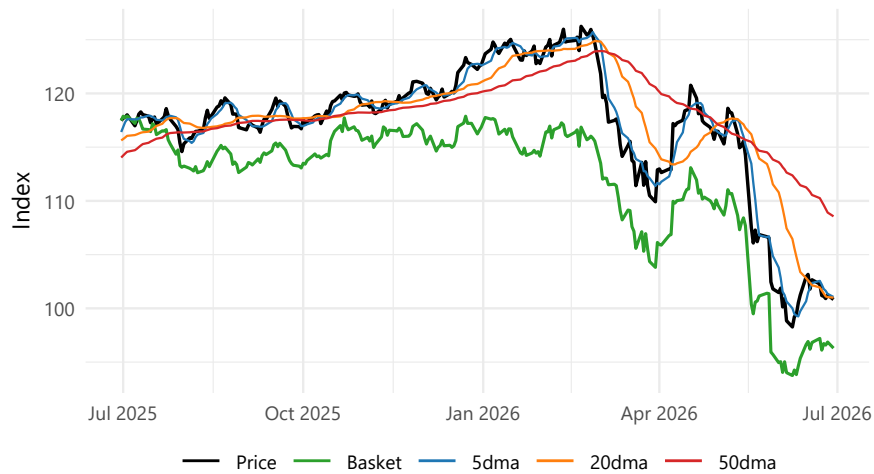
	ticker	PC_wt	name	category	etf_signal
2	DBC	0.1055	Invesco DB Commodity Index Tracking Fund	Broad Commodities	LONG
6	GSG	0.1055	iShares GSCI Commodity-Indexed Trust Fu...	Broad Commodities	FLAT
8	PDBC	0.1055	Invesco Optimum Yield Diversified Commo...	Broad Commodities	LONG
9	COMT	0.1052	iShares GSCI Commodity Dynamic Roll Str...	Broad Commodities	FLAT
10	DBE	0.1019	Invesco DB Energy Fund	Energy Commodities	LONG
1	USO	0.0995	United States Oil Fund	Oil	LONG
5	DBO	0.0985	Invesco DB Oil Fund	Oil	FLAT
7	DJP	0.0982	iPath Bloomberg Commodity Index Total R...	Broad Commodities	LONG
4	FTGC	0.0973	First Trust Global Tactical Commodity S...	Broad Commodities	FLAT
3	DBA	0.0499	Invesco DB Agriculture Fund	Agricultural Commodities	FLAT
11	UNG	0.0329	United States Natural Gas Fund LP	Natural Gas	FLAT

Cluster 20 – India Equity

Members: 10 **Var explained:** 0.405 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +4.7%)
- Latest sell_cross: 2026-05-13
- Current 252d return: -13.6% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-05-13 more recent than last effective buy 2026-05-06)



Category breakdown

category	n_members	abs_weight
India Equity	3	0.4677
Malaysia Equity	1	0.1125
Indonesia Equity	1	0.1042
Philippines Equity	1	0.1009
Vietnam Equity	1	0.0752
Turkey Equity	1	0.0611
U.S. Dividend Equity	1	0.0548
Platinum	1	0.0235

Replication basket – NOBL+INDA+PPLT (cor=0.92) basket lead at lag 1: -0.007

ticker	rep_wt	name	category	etf_signal
NOBL	0.080	ProShares S&P 500 Dividend Aristocrat...	U.S. Dividend Equity	FLAT
INDA	0.660	iShares MSCI India ETF	India Equity	FLAT
PPLT	0.029	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	INDA	0.1573	iShares MSCI India ETF	India Equity	FLAT
2	INDY	0.1556	iShares India 50 ETF	India Equity	FLAT
3	EPI	0.1549	WisdomTree India Earnings Fund	India Equity	FLAT
8	EWM	0.1125	iShares MSCI Malaysia Index Fund	Malaysia Equity	FLAT
6	EIDO	0.1042	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
7	EPHE	0.1009	iShares MSCI Philippines ETF	Philippines Equity	FLAT
10	VNM	0.0752	VanEck Vietnam ETF	Vietnam Equity	FLAT
9	TUR	0.0611	iShares MSCI Turkey ETF	Turkey Equity	FLAT
4	NOBL	0.0548	ProShares S&P 500 Dividend Aristocrats ...	U.S. Dividend Equity	FLAT
5	PPLT	0.0235	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Cluster 21 – Inflation-Protected Fixed Income

Members: 10 **Var explained:** 0.906 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-27 – filter **PASS** (252d at date = +4.8%)
- Latest sell_cross: 2026-06-03
- Current 252d return: +2.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-03 more recent than last effective buy 2026-05-27)



Category breakdown

category	n_members	abs_weight
Inflation-Protected Fixed Income	10	1

Replication basket – VTIP+SCHP (cor=0.99) basket lead at lag 1: -0.015

ticker	rep_wt	name	category	etf_signal
VTIP	0.262	Vanguard Short-Term Inflation-Protect...	Inflation-Protected F...	FLAT
SCHP	0.762	Schwab U.S. TIPS ETF	Inflation-Protected F...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	TDTF	0.1035	FlexShares iBoxx 5 Year Target Duration...	Inflation-Protected Fixe...	LONG
6	TIPX	0.1034	State Street SPDR Bloomberg 1a??10 Year...	Inflation-Protected Fixe...	FLAT
3	TIP	0.1030	iShares TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
8	SCHP	0.1023	Schwab U.S. TIPS ETF	Inflation-Protected Fixe...	FLAT
5	TDTT	0.1007	FlexShares iBoxx 3 Year Target Duration...	Inflation-Protected Fixe...	FLAT
7	SPIP	0.0999	State Street SPDR Portfolio TIPS ETF	Inflation-Protected Fixe...	FLAT

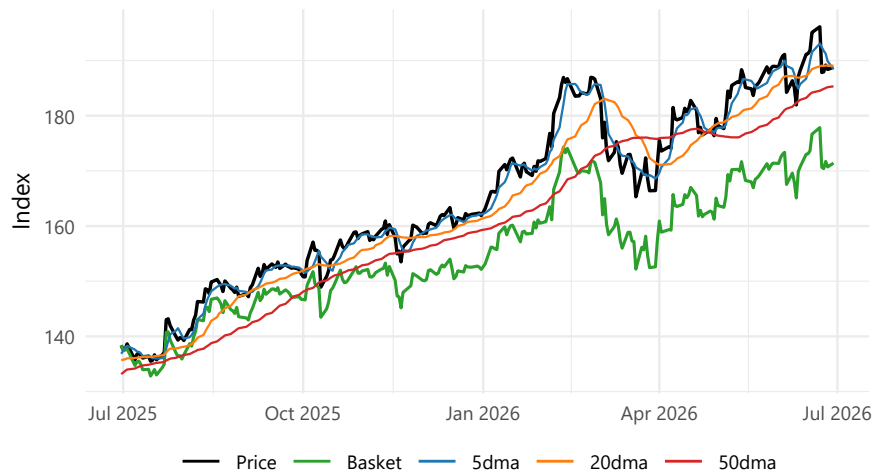
9	STIP	0.0998	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
2	STPZ	0.0996	PIMCO 1-5 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT
10	VTIP	0.0994	Vanguard Short-Term Inflation-Protected...	Inflation-Protected Fixe...	FLAT
1	LTPZ	0.0885	Pimco 15 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	LONG

Cluster 22 – Japan Equity

Members: 9 **Var explained:** 0.888 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-09 – filter **PASS** (252d at date = +57.2%)
- Latest sell_cross: 2026-06-11
- Current 252d return: +37.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-11 more recent than last effective buy 2026-04-09)



Category breakdown

category	n_members	abs_weight
Japan Equity	7	0.7803

Asia-Pacific Equity	2	0.2197
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Replication basket – EWJ (cor=0.97) basket lead at lag 1: -0.041

ticker	rep_wt	name	category	etf_signal
EWJ	0.909	iShares MSCI Japan Index Fund	Japan Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

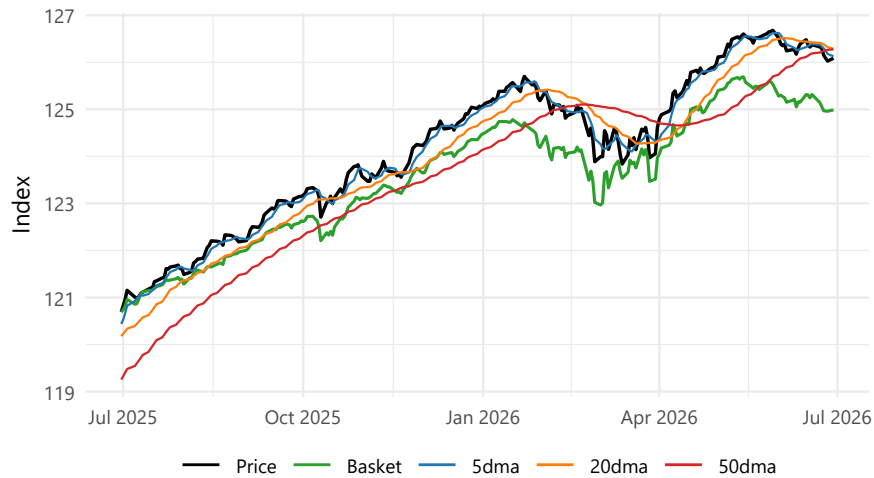
	ticker	PC_wt	name	category	etf_signal
1	EWJ	0.1142	iShares MSCI Japan Index Fund	Japan Equity	FLAT
8	BBJP	0.1142	JPMorgan BetaBuilders Japan ETF	Japan Equity	FLAT
6	DBJP	0.1117	Xtrackers MSCI Japan Hedged Equity ETF	Japan Equity	LONG
9	HEWJ	0.1117	iShares Currency Hedged MSCI Japan ETF	Japan Equity	LONG
3	IPAC	0.1113	iShares Core MSCI Pacific ETF	Asia-Pacific Equity	FLAT
5	FLJH	0.1109	Franklin FTSE Japan Hedged ETF	Japan Equity	FLAT
2	DXJ	0.1096	WisdomTree Japan Hedged Equity Fund	Japan Equity	FLAT
4	VPL	0.1084	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	FLAT
7	EWJV	0.1080	iShares MSCI Japan Value ETF	Japan Equity	FLAT

Cluster 23 – Bank Loan / Floating Rate

Members: 9 **Var explained:** 0.558 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +6.5%)
- Latest sell_cross: 2026-06-23
- Current 252d return: +4.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-23 more recent than last effective buy 2026-04-07)



Category breakdown

category	n_members	abs_weight
Bank Loan / Floating Rate	3	0.3626
Interest Rate Hedged Fixed Income	3	0.3246
Floating Rate Fixed Income	3	0.3129

Replication basket – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.047

ticker	rep_wt	name	category	etf_signal
FLOT	0.293	iShares Floating Rate Bond ETF	Floating Rate Fixed I...	FLAT
BKLN	0.208	Invesco Senior Loan ETF	Bank Loan / Floating ...	FLAT
SRLN	0.310	State Street Blackstone Senior Loan E...	Bank Loan / Floating ...	FLAT
FLRN	0.128	State Street SPDR Bloomberg Investmen...	Floating Rate Fixed I...	FLAT
FLTR	0.034	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	SRLN	0.1250	State Street Blackstone Senior Loan ETF	Bank Loan / Floating Rate	FLAT

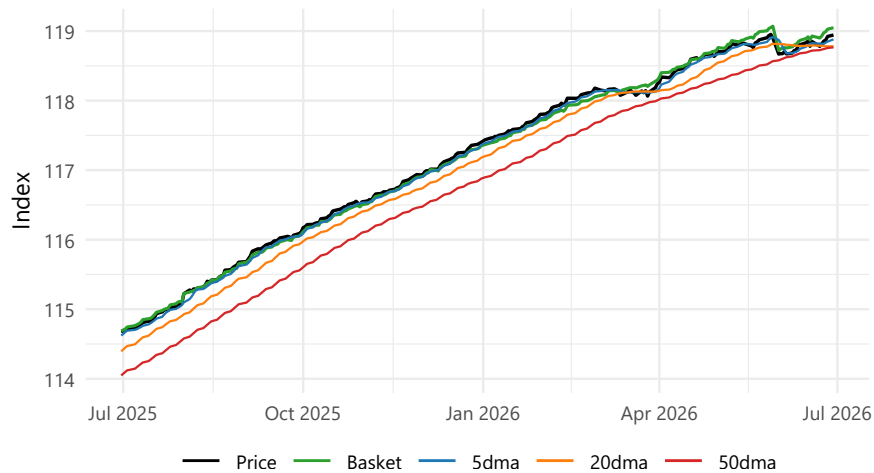
3	BKLN	0.1210	Invesco Senior Loan ETF	Bank Loan / Floating Rate	FLAT
1	FTSL	0.1166	First Trust Senior Loan Fund	Bank Loan / Floating Rate	FLAT
5	LQDH	0.1145	iShares Interest Rate Hedged Corporate ...	Interest Rate Hedged Fix...	FLAT
7	FLOT	0.1082	iShares Floating Rate Bond ETF	Floating Rate Fixed Inco...	FLAT
6	IGBH	0.1057	iShares Interest Rate Hedged Long-Term ...	Interest Rate Hedged Fix...	FLAT
4	IGHG	0.1043	ProShares Investment Grade-Interest Rat...	Interest Rate Hedged Fix...	FLAT
8	FLRN	0.1037	State Street SPDR Bloomberg Investment ...	Floating Rate Fixed Inco...	FLAT
9	FLTR	0.1009	VanEck IG Floating Rate ETF	Floating Rate Fixed Inco...	FLAT

Cluster 24 – Cash / Ultra-Short Fixed Income

Members: 9 Var explained: 0.453 Signal: FLAT

Signal logic:

- Latest buy_cross: 2022-10-28 – filter **BLOCKED** (252d at date = -0.5%; needs > 0)
- Latest effective buy: 2022-11-29 – delayed re-arm from blocked cross (252d at fire = +0.0%)
- Latest sell_cross: 2026-06-03
- Current 252d return: +3.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-03 more recent than last effective buy 2022-11-29)



Category breakdown

category	n_members	abs_weight
Cash / Ultra-Short Fixed Income	9	1

Replication basket – SHV+MINT+GBIL+ICSH (cor=0.84) basket lead at lag 1: 0.088

ticker	rep_wt	name	category	etf_signal
SHV	0.297	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fi...	FLAT
MINT	0.177	PIMCO Enhanced Short Maturity Active ...	Cash / Ultra-Short Fi...	FLAT
GBIL	0.121	Goldman Sachs Access Treasury 0-1 Yea...	Cash / Ultra-Short Fi...	FLAT
ICSH	0.413	iShares Ultra Short Duration Bond Act...	Cash / Ultra-Short Fi...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

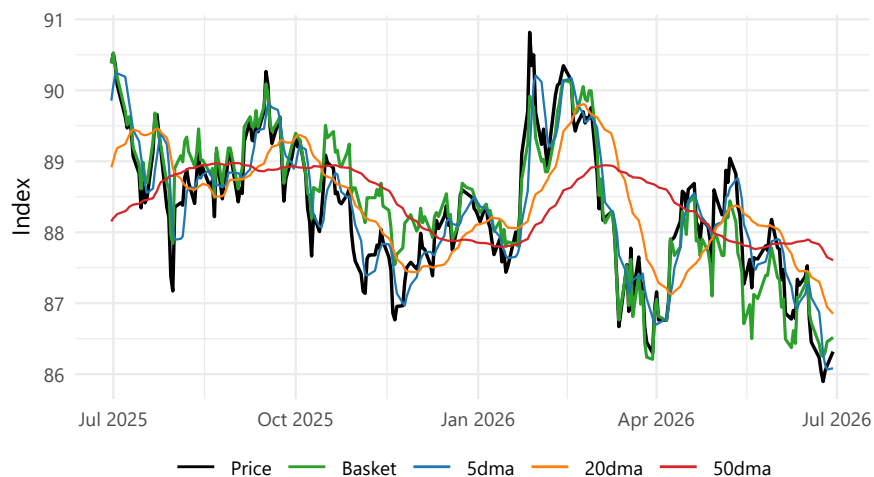
	ticker	PC_wt	name	category	etf_signal
7	SHV	0.1284	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fixed...	FLAT
6	ICSH	0.1277	iShares Ultra Short Duration Bond Activ...	Cash / Ultra-Short Fixed...	FLAT
5	GSY	0.1251	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fixed...	FLAT
3	FTSM	0.1204	First Trust Enhanced Short Maturity ETF	Cash / Ultra-Short Fixed...	FLAT
4	NEAR	0.1104	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fixed...	FLAT
2	MINT	0.1097	PIMCO Enhanced Short Maturity Active Ex...	Cash / Ultra-Short Fixed...	FLAT
1	VNLA	0.0989	Janus Henderson Short Duration Income E...	Cash / Ultra-Short Fixed...	FLAT
8	BIL	0.0950	State Street SPDR Bloomberg 1-3 Month T...	Cash / Ultra-Short Fixed...	FLAT
9	GBIL	0.0843	Goldman Sachs Access Treasury 0-1 Year ...	Cash / Ultra-Short Fixed...	FLAT

Cluster 25 – International Fixed Income

Members: 6 **Var explained:** 0.742 **Signal:** LONG since 2026-05-29 (20 days, -2.1%)

Signal logic:

- Latest buy_cross: 2026-05-29 – filter **PASS** (252d at date = +0.6%)
- Latest sell_cross: 2026-05-18
- Current 252d return: -4.1% – filter **BLOCKS** new entries today
- Decision: **LONG** (effective buy at 2026-05-29 > last sell at 2026-05-18)



Category breakdown

category	n_members	abs_weight
Currency	4	0.6409
International Fixed Income	2	0.3591

Replication basket – BWX (cor=0.93) basket lead at lag 1: 0.027

ticker	rep_wt	name	category	etf_signal
BWX	0.751	State Street SPDR Bloomberg Internati...	International Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	IGOV	0.1801	iShares International Treasury Bond ETF	International Fixed Inco...	FLAT
1	BWX	0.1790	State Street SPDR Bloomberg Internation...	International Fixed Inco...	FLAT
4	FXE	0.1702	Invesco CurrencyShares Euro Currency Tr...	Currency	FLAT
5	FXF	0.1649	Invesco CurrencyShares Swiss Franc Trust	Currency	FLAT
3	FXB	0.1610	Invesco CurrencyShares British Pound St...	Currency	FLAT
6	FXY	0.1448	Invesco CurrencyShares Japanese Yen Tru...	Currency	FLAT

Cluster 26 – Leveraged / Inverse**Members:** 2 **Var explained:** 0.626 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-02-05 – filter **BLOCKED** (252d at date = -32.0%; needs > 0)
- Latest effective buy: 2025-04-08 – delayed re-arm from blocked cross (252d at fire = +0.4%)
- Latest sell_cross: 2026-06-18
- Current 252d return: -36.8% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-06-18 more recent than last effective buy 2025-04-08)

**Category breakdown**

category	n_members	abs_weight
Leveraged / Inverse	1	0.5
Currency	1	0.5

Replication basket – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.024

ticker	rep_wt	name	category	etf_signal
SQQQ	0.506	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT

UUP	0.495	Invesco DB USD Index Bullish Fund ETF	Currency	LONG
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Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
SQQQ	0.5	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.5	Invesco DB USD Index Bullish Fund ETF	Currency	LONG

Long ETF Positions (full)

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

ticker	cluster	theme	entry	days	return	p_sell	nss
PSCH	cluster_11	Healthcare	2026-04-13	53	22.0%	0.00	1.00
IBB	cluster_11	Healthcare	2026-06-17	7	9.4%	0.00	1.00
XBI	cluster_11	Healthcare	2026-06-16	8	17.0%	0.00	1.00
PTH	cluster_11	Healthcare	2026-06-17	7	12.9%	0.00	1.00
IHF	cluster_11	Healthcare	2026-05-08	34	10.7%	0.00	1.00
FXH	cluster_11	Healthcare	2026-04-10	54	9.9%	0.00	1.00
VHT	cluster_11	Healthcare	2026-05-19	27	10.4%	0.00	1.00
XHE	cluster_11	Healthcare	2026-06-09	13	3.5%	0.00	1.00
FHLC	cluster_11	Healthcare	2026-05-19	27	10.3%	0.00	1.00
RSPH	cluster_11	Healthcare	2026-05-21	25	8.6%	0.00	1.00
IYH	cluster_11	Healthcare	2026-05-20	26	9.8%	0.00	1.00
XLV	cluster_11	Healthcare	2026-05-21	25	8.5%	0.01	0.99
PPH	cluster_11	Healthcare	2026-05-21	25	6.4%	0.03	0.97
EDV	cluster_13	Fixed Income	2026-06-12	10	3.1%	0.00	1.00
TLT	cluster_13	Fixed Income	2026-06-16	8	1.5%	0.00	1.00
SPTL	cluster_13	Fixed Income	2026-06-16	8	1.3%	0.00	1.00
VGLT	cluster_13	Fixed Income	2026-06-16	8	1.3%	0.00	1.00
TLH	cluster_13	Fixed Income	2026-06-16	8	1.1%	0.00	1.00
BLV	cluster_13	Fixed Income	2026-06-15	9	1.4%	0.00	1.00
BAB	cluster_13	Fixed Income	2026-06-16	8	0.4%	0.00	1.00
IGLB	cluster_13	Fixed Income	2026-06-12	10	1.0%	0.00	1.00
SPLB	cluster_13	Fixed Income	2026-06-12	10	1.0%	0.00	1.00
VCLT	cluster_13	Fixed Income	2026-06-12	10	1.0%	0.00	1.00
FLCO	cluster_13	Fixed Income	2026-06-16	8	0.4%	0.00	0.99
LQD	cluster_13	Fixed Income	2026-06-16	8	0.5%	0.01	0.98
USIG	cluster_13	Fixed Income	2026-06-16	8	0.5%	0.01	0.98
QLTA	cluster_13	Fixed Income	2026-06-17	7	0.8%	0.01	0.98
CORP	cluster_13	Fixed Income	2026-06-16	8	0.5%	0.00	0.98
XHB	cluster_1	U.S. Mid Cap Equity	2026-05-29	20	12.4%	0.00	1.00
ITB	cluster_1	U.S. Mid Cap Equity	2026-06-09	13	8.0%	0.00	1.00
PSCD	cluster_1	U.S. Mid Cap Equity	2026-05-27	22	8.5%	0.00	1.00
PKB	cluster_1	U.S. Mid Cap Equity	2026-06-16	8	2.9%	0.00	1.00
FIW	cluster_1	U.S. Mid Cap Equity	2026-06-16	8	1.5%	0.00	1.00
PHO	cluster_1	U.S. Mid Cap Equity	2026-06-26	1	-1.0%	0.00	1.00
SDVY	cluster_1	U.S. Mid Cap Equity	2026-06-08	14	3.8%	0.00	1.00
PEJ	cluster_1	U.S. Mid Cap Equity	2026-05-26	23	9.3%	0.00	1.00
CGW	cluster_1	U.S. Mid Cap Equity	2026-06-25	2	0.3%	0.00	1.00

FXR	cluster_1	U.S. Mid Cap Equity	2026-06-12	10	3.1%	0.00	1.00
ISMD	cluster_1	U.S. Mid Cap Equity	2026-04-09	55	17.4%	0.00	1.00
SCHA	cluster_1	U.S. Mid Cap Equity	2026-04-09	55	15.7%	0.00	1.00
XRT	cluster_1	U.S. Mid Cap Equity	2026-05-28	21	2.7%	0.00	1.00
IJS	cluster_1	U.S. Mid Cap Equity	2026-04-10	54	11.0%	0.01	0.99
VFMO	cluster_1	U.S. Mid Cap Equity	2026-04-08	56	17.4%	0.01	0.99
IWO	cluster_1	U.S. Mid Cap Equity	2026-04-13	53	15.2%	0.02	0.98
VTWG	cluster_1	U.S. Mid Cap Equity	2026-04-13	53	15.2%	0.02	0.98
PDP	cluster_1	U.S. Mid Cap Equity	2026-04-08	56	15.8%	0.03	0.97
FXD	cluster_1	U.S. Mid Cap Equity	2026-05-27	22	2.5%	0.05	0.95
SCHM	cluster_1	U.S. Mid Cap Equity	2026-04-09	55	12.2%	0.08	0.92
IMCG	cluster_1	U.S. Mid Cap Equity	2026-04-09	55	16.9%	0.12	0.88
MTUM	cluster_1	U.S. Mid Cap Equity	2026-04-08	56	29.4%	0.16	0.84
ITA	cluster_1	U.S. Mid Cap Equity	2026-05-26	23	4.2%	0.16	0.84
RSPD	cluster_1	U.S. Mid Cap Equity	2026-06-11	11	0.8%	0.20	0.80
IVOG	cluster_1	U.S. Mid Cap Equity	2026-04-09	55	9.8%	0.23	0.77
IJK	cluster_1	U.S. Mid Cap Equity	2026-04-09	55	9.8%	0.25	0.74
FTC	cluster_1	U.S. Mid Cap Equity	2026-04-10	54	15.7%	0.26	0.74
RPG	cluster_1	U.S. Mid Cap Equity	2026-04-09	55	22.5%	0.26	0.73
MDYG	cluster_1	U.S. Mid Cap Equity	2026-04-09	55	9.8%	0.28	0.71
VOT	cluster_1	U.S. Mid Cap Equity	2026-04-14	52	9.8%	0.48	0.49
FV	cluster_1	U.S. Mid Cap Equity	2026-04-14	52	16.2%	0.59	0.41
FTXR	cluster_1	U.S. Mid Cap Equity	2026-04-17	49	9.6%	0.66	0.34
IWP	cluster_1	U.S. Mid Cap Equity	2026-04-15	51	6.9%	0.61	0.30
XMMO	cluster_1	U.S. Mid Cap Equity	2026-04-06	58	13.5%	0.64	0.18
PPA	cluster_1	U.S. Mid Cap Equity	2026-05-27	22	-0.8%	0.95	-0.39
FXO	cluster_14	Financials	2026-06-05	15	5.0%	0.00	1.00
IAT	cluster_14	Financials	2026-05-20	26	9.6%	0.00	1.00
KIE	cluster_14	Financials	2026-06-10	12	5.7%	0.00	1.00
FNCL	cluster_14	Financials	2026-04-13	53	4.2%	0.00	1.00
XLFF	cluster_14	Financials	2026-04-13	53	4.0%	0.00	1.00
HYMB	cluster_16	Municipal Bond	2026-05-26	23	1.6%	0.00	1.00
SHM	cluster_16	Municipal Bond	2026-05-28	21	0.2%	0.00	1.00
MUNI	cluster_16	Municipal Bond	2026-05-28	21	0.4%	0.00	1.00
MUB	cluster_16	Municipal Bond	2026-05-28	21	0.6%	0.00	1.00
CMF	cluster_16	Municipal Bond	2026-05-28	21	0.7%	0.00	1.00
NYF	cluster_16	Municipal Bond	2026-05-28	21	0.7%	0.00	1.00
VTEB	cluster_16	Municipal Bond	2026-05-28	21	0.7%	0.00	1.00
TFI	cluster_16	Municipal Bond	2026-05-28	21	0.6%	0.00	1.00
ITM	cluster_16	Municipal Bond	2026-05-28	21	0.4%	0.01	0.99
HYD	cluster_16	Municipal Bond	2026-05-22	24	1.8%	0.08	0.92
FMB	cluster_16	Municipal Bond	2026-05-29	20	0.5%	0.10	0.90
PZA	cluster_16	Municipal Bond	2026-05-29	20	1.0%	0.11	0.89
SUB	cluster_16	Municipal Bond	2026-05-27	22	0.1%	0.95	-0.23
IAGG	cluster_6	Aggregate Fixed Income	2026-05-22	24	1.1%	0.00	1.00

BNDX	cluster_6	Aggregate Fixed Income	2026-05-26	23	0.6%	0.00	1.00
GOVT	cluster_6	Aggregate Fixed Income	2026-06-25	2	0.2%	0.00	1.00
FBND	cluster_6	Aggregate Fixed Income	2026-06-17	7	0.4%	0.00	1.00
BOND	cluster_6	Aggregate Fixed Income	2026-06-17	7	1.0%	0.00	0.99
SPMB	cluster_6	Aggregate Fixed Income	2026-06-16	8	0.4%	0.00	0.99
BND	cluster_6	Aggregate Fixed Income	2026-06-24	3	0.2%	0.00	0.99
SPAB	cluster_6	Aggregate Fixed Income	2026-06-24	3	0.2%	0.00	0.99
AGG	cluster_6	Aggregate Fixed Income	2026-06-24	3	0.2%	0.00	0.99
IUSB	cluster_6	Aggregate Fixed Income	2026-06-17	7	0.7%	0.00	0.99
AGGY	cluster_6	Aggregate Fixed Income	2026-06-16	8	0.1%	0.00	0.99
GNMA	cluster_6	Aggregate Fixed Income	2026-06-25	2	0.1%	0.00	0.99
SCHZ	cluster_6	Aggregate Fixed Income	2026-06-24	3	0.2%	0.00	0.99
IEF	cluster_6	Aggregate Fixed Income	2026-06-26	1	0.0%	0.00	0.98
VMBS	cluster_6	Aggregate Fixed Income	2026-06-24	3	0.2%	0.00	0.98
MBB	cluster_6	Aggregate Fixed Income	2026-06-24	3	0.2%	0.00	0.97
AGZ	cluster_6	Aggregate Fixed Income	2026-06-26	1	-0.0%	0.00	0.96
BIV	cluster_6	Aggregate Fixed Income	2026-06-26	1	0.1%	0.00	0.93
VCIT	cluster_6	Aggregate Fixed Income	2026-06-25	2	0.2%	0.01	0.90
IGIB	cluster_6	Aggregate Fixed Income	2026-06-25	2	0.2%	0.01	0.89
LDUR	cluster_6	Aggregate Fixed Income	2026-06-25	2	0.1%	0.01	0.84
SPTI	cluster_6	Aggregate Fixed Income	2026-06-29	0	0.0%	0.00	0.83
VGIT	cluster_6	Aggregate Fixed Income	2026-06-29	0	0.0%	0.00	0.83
SPIB	cluster_6	Aggregate Fixed Income	2026-06-26	1	0.1%	0.02	0.82
SCHR	cluster_6	Aggregate Fixed Income	2026-06-29	0	0.0%	0.01	0.79
GVI	cluster_6	Aggregate Fixed Income	2026-06-29	0	0.0%	0.01	0.70
IEI	cluster_6	Aggregate Fixed Income	2026-06-29	0	0.0%	0.01	0.67
VCSH	cluster_6	Aggregate Fixed Income	2026-06-26	1	0.0%	0.04	0.62
CMBS	cluster_6	Aggregate Fixed Income	2026-06-29	0	0.0%	0.03	0.59
SLQD	cluster_6	Aggregate Fixed Income	2026-06-26	1	0.0%	0.05	0.57
IGSB	cluster_6	Aggregate Fixed Income	2026-06-29	0	0.0%	0.06	0.53
SPSB	cluster_6	Aggregate Fixed Income	2026-06-26	1	0.0%	0.07	0.52
GTO	cluster_6	Aggregate Fixed Income	2026-06-29	0	0.0%	0.24	0.52
TOTL	cluster_6	Aggregate Fixed Income	2026-06-29	0	0.0%	0.05	0.51
LMBS	cluster_6	Aggregate Fixed Income	2026-06-16	8	-0.3%	0.08	0.46
ISTB	cluster_6	Aggregate Fixed Income	2026-06-29	0	0.0%	0.06	0.39
PSCC	cluster_18	Consumer Staples	2026-06-10	12	4.3%	0.00	1.00
IDU	cluster_18	Consumer Staples	2026-06-25	2	0.3%	0.00	1.00
XLU	cluster_18	Consumer Staples	2026-06-25	2	0.4%	0.00	0.99
VPU	cluster_18	Consumer Staples	2026-06-25	2	0.3%	0.00	0.99
FUTY	cluster_18	Consumer Staples	2026-06-25	2	0.3%	0.00	0.98
IYK	cluster_18	Consumer Staples	2026-06-09	13	1.4%	0.09	0.85
VDC	cluster_18	Consumer Staples	2026-06-29	0	0.0%	0.12	0.41
FSTA	cluster_18	Consumer Staples	2026-06-29	0	0.0%	0.14	0.36
XLP	cluster_18	Consumer Staples	2026-06-29	0	0.0%	0.25	0.33
KXI	cluster_18	Consumer Staples	2026-06-29	0	0.0%	0.17	0.31

NURE	cluster_15	Real Estate	2026-04-13	53	13.8%	0.02	0.98
RWO	cluster_15	Real Estate	2026-06-25	2	1.4%	0.05	0.79
INDS	cluster_15	Real Estate	2026-06-08	14	3.3%	0.09	0.78
IYR	cluster_15	Real Estate	2026-06-25	2	1.0%	0.07	0.76
FREL	cluster_15	Real Estate	2026-06-25	2	1.0%	0.13	0.72
XLRE	cluster_15	Real Estate	2026-06-25	2	0.7%	0.20	0.55
REGL	cluster_2	U.S. Large Cap Value	2026-06-12	10	1.5%	0.00	1.00
MGV	cluster_2	U.S. Large Cap Value	2026-04-13	53	9.4%	0.00	1.00
PKW	cluster_2	U.S. Large Cap Value	2026-05-20	26	4.5%	0.00	1.00
LVHD	cluster_2	U.S. Large Cap Value	2026-06-10	12	2.2%	0.00	1.00
VTV	cluster_2	U.S. Large Cap Value	2026-04-13	53	7.9%	0.00	1.00
AIVL	cluster_2	U.S. Large Cap Value	2026-04-10	54	8.6%	0.00	1.00
SPLV	cluster_2	U.S. Large Cap Value	2026-06-11	11	2.4%	0.00	1.00
PWV	cluster_2	U.S. Large Cap Value	2026-04-02	59	9.9%	0.01	0.99
SMMV	cluster_2	U.S. Large Cap Value	2026-06-25	2	2.1%	0.00	0.99
XMLV	cluster_2	U.S. Large Cap Value	2026-06-25	2	0.8%	0.00	0.97
FVD	cluster_2	U.S. Large Cap Value	2026-06-10	12	1.4%	0.03	0.96
SPHD	cluster_2	U.S. Large Cap Value	2026-06-08	14	3.4%	0.04	0.96
FEX	cluster_2	U.S. Large Cap Value	2026-04-09	55	9.8%	0.05	0.95
IMCB	cluster_2	U.S. Large Cap Value	2026-04-10	54	11.4%	0.05	0.95
IWR	cluster_2	U.S. Large Cap Value	2026-04-10	54	9.5%	0.06	0.94
DIA	cluster_2	U.S. Large Cap Value	2026-04-14	52	7.5%	0.06	0.94
FTA	cluster_2	U.S. Large Cap Value	2026-05-21	25	3.6%	0.07	0.93
SCHV	cluster_2	U.S. Large Cap Value	2026-04-10	54	10.3%	0.07	0.93
SDY	cluster_2	U.S. Large Cap Value	2026-05-20	26	3.1%	0.08	0.92
IWD	cluster_2	U.S. Large Cap Value	2026-04-10	54	9.4%	0.10	0.90
LGLV	cluster_2	U.S. Large Cap Value	2026-06-26	1	-0.2%	0.04	0.89
VONV	cluster_2	U.S. Large Cap Value	2026-04-10	54	9.4%	0.16	0.84
VO	cluster_2	U.S. Large Cap Value	2026-04-13	53	7.3%	0.20	0.80
IWX	cluster_2	U.S. Large Cap Value	2026-04-10	54	9.6%	0.30	0.70
HDV	cluster_2	U.S. Large Cap Value	2026-06-25	2	0.6%	0.23	0.62
RSP	cluster_2	U.S. Large Cap Value	2026-04-14	52	6.7%	0.39	0.61
DGRO	cluster_2	U.S. Large Cap Value	2026-04-13	53	5.1%	0.47	0.53
SIZE	cluster_2	U.S. Large Cap Value	2026-04-13	53	7.1%	0.52	0.47
DTD	cluster_2	U.S. Large Cap Value	2026-04-13	53	5.1%	0.63	0.35
DIV	cluster_2	U.S. Large Cap Value	2026-06-29	0	0.0%	0.12	0.28
VLUE	cluster_2	U.S. Large Cap Value	2026-04-09	55	31.4%	0.75	0.25
FTCS	cluster_2	U.S. Large Cap Value	2026-06-26	1	-0.1%	0.44	0.22
PRF	cluster_2	U.S. Large Cap Value	2026-04-09	55	9.2%	0.76	0.21
JVAL	cluster_2	U.S. Large Cap Value	2026-04-10	54	14.9%	0.81	0.18
VIG	cluster_2	U.S. Large Cap Value	2026-04-14	52	5.4%	0.85	0.13
RWL	cluster_2	U.S. Large Cap Value	2026-04-10	54	8.1%	0.87	0.12
FNDX	cluster_2	U.S. Large Cap Value	2026-04-09	55	8.1%	0.91	0.01
DLN	cluster_2	U.S. Large Cap Value	2026-04-13	53	5.0%	0.88	-0.05
VOOV	cluster_2	U.S. Large Cap Value	2026-04-14	52	4.4%	0.99	-0.72

USMV	cluster_2	U.S. Large Cap Value	2026-06-29	0	0.0%	0.92	-0.73
SCHD	cluster_2	U.S. Large Cap Value	2026-06-26	1	-0.5%	1.00	-0.98
UUP	cluster_26	Leveraged / Inverse	2026-05-18	28	2.4%	0.00	1.00
HEWJ	cluster_22	Japan Equity	2026-04-08	56	7.6%	0.37	0.62
DBJP	cluster_22	Japan Equity	2026-04-08	56	6.4%	0.62	0.34
LTPZ	cluster_21	Inflation-Protected Fix...	2026-06-26	1	0.4%	0.06	0.65
TDTF	cluster_21	Inflation-Protected Fix...	2026-05-28	21	-1.4%	0.62	-0.62
IBUY	cluster_3	U.S. Large Cap Growth	2026-06-15	9	2.2%	0.00	1.00
PSI	cluster_3	U.S. Large Cap Growth	2026-04-07	57	78.8%	0.06	0.94
EWS	cluster_3	U.S. Large Cap Growth	2026-06-17	7	0.8%	0.17	0.77
SPHB	cluster_3	U.S. Large Cap Growth	2026-04-10	54	25.3%	0.41	0.58
SOXX	cluster_3	U.S. Large Cap Growth	2026-04-08	56	65.9%	0.46	0.54
FTXL	cluster_3	U.S. Large Cap Growth	2026-04-07	57	75.0%	0.50	0.50
IPO	cluster_3	U.S. Large Cap Growth	2026-04-13	53	27.2%	0.54	0.46
SMH	cluster_3	U.S. Large Cap Growth	2026-04-08	56	49.4%	0.58	0.41
PSCT	cluster_3	U.S. Large Cap Growth	2026-04-07	57	42.7%	0.69	0.29
XNTK	cluster_3	U.S. Large Cap Growth	2026-04-10	54	37.4%	0.81	0.17
QTEC	cluster_3	U.S. Large Cap Growth	2026-04-10	54	44.2%	0.81	0.16
QQEW	cluster_3	U.S. Large Cap Growth	2026-04-16	50	17.0%	0.85	0.10
QTUM	cluster_3	U.S. Large Cap Growth	2026-04-10	54	37.8%	0.92	0.03
PTF	cluster_3	U.S. Large Cap Growth	2026-04-07	57	41.4%	0.81	-0.01
ARKK	cluster_3	U.S. Large Cap Growth	2026-06-29	0	0.0%	0.56	-0.25
XSD	cluster_3	U.S. Large Cap Growth	2026-04-09	55	62.7%	0.98	-0.45
ARKQ	cluster_3	U.S. Large Cap Growth	2026-06-17	7	-2.3%	1.00	-1.00
EWT	cluster_7	Emerging Markets Equity	2026-04-08	56	38.5%	0.72	0.26
EWX	cluster_7	Emerging Markets Equity	2026-06-15	9	-3.0%	0.85	-0.75
FALN	cluster_10	High Yield Bond	2026-06-12	10	0.5%	0.00	0.98
ANGL	cluster_10	High Yield Bond	2026-06-15	9	0.2%	0.01	0.93
VWOB	cluster_10	High Yield Bond	2026-05-26	23	1.3%	0.07	0.91
PCY	cluster_10	High Yield Bond	2026-05-28	21	1.1%	0.33	0.62
HYEM	cluster_10	High Yield Bond	2026-04-13	53	1.1%	0.49	0.14
HYS	cluster_10	High Yield Bond	2026-06-16	8	0.0%	0.29	-0.02
BBHY	cluster_10	High Yield Bond	2026-06-25	2	0.2%	0.32	-0.08
JNK	cluster_10	High Yield Bond	2026-06-16	8	-0.0%	0.36	-0.11
SHYG	cluster_10	High Yield Bond	2026-06-25	2	0.1%	0.29	-0.13
SJNK	cluster_10	High Yield Bond	2026-06-17	7	0.2%	0.32	-0.17
HYG	cluster_10	High Yield Bond	2026-06-25	2	0.2%	0.41	-0.24
HYLB	cluster_10	High Yield Bond	2026-06-17	7	0.3%	0.38	-0.25
IGF	cluster_8	Asia-Pacific Equity	2026-06-29	0	0.0%	0.01	0.58
EWL	cluster_8	Asia-Pacific Equity	2026-06-25	2	0.7%	0.17	0.51
EMLC	cluster_8	Asia-Pacific Equity	2026-06-16	8	-0.9%	0.49	-0.13
EBND	cluster_8	Asia-Pacific Equity	2026-06-17	7	-0.2%	0.50	-0.26
IHDG	cluster_4	Developed ex-US Equity	2026-05-06	36	2.6%	0.28	0.72
EWO	cluster_4	Developed ex-US Equity	2026-04-10	54	8.5%	0.28	0.72
EUFN	cluster_4	Developed ex-US Equity	2026-06-12	10	-0.3%	0.35	0.35

EWN	cluster_4	Developed ex-US Equity	2026-04-10	54	10.6%	0.88	0.08
IQLT	cluster_4	Developed ex-US Equity	2026-06-12	10	-1.2%	0.94	-0.93
SPHQ	cluster_5	U.S. Large Cap Equity	2026-04-10	54	12.7%	0.40	0.60
ASHR	cluster_12	China Equity	2026-06-15	9	0.3%	0.12	0.67
CNYA	cluster_12	China Equity	2026-06-15	9	-0.2%	0.10	0.64
SIVR	cluster_17	Gold	2026-05-27	22	-22.0%	1.00	-1.00
SLV	cluster_17	Gold	2026-05-27	22	-22.0%	1.00	-1.00
EMLP	cluster_9	Materials	2026-06-26	1	-0.5%	0.01	0.80
MLPX	cluster_9	Materials	2026-06-26	1	-1.3%	0.06	0.68
XLB	cluster_9	Materials	2026-06-16	8	-3.9%	0.61	-0.22
RSPM	cluster_9	Materials	2026-06-16	8	-3.9%	0.72	-0.51
VAW	cluster_9	Materials	2026-06-16	8	-4.2%	0.78	-0.58
MLPA	cluster_9	Materials	2026-06-08	14	-2.1%	0.92	-0.92
DBE	cluster_19	Broad Commodities	2026-06-03	17	-18.4%	1.00	-1.00
USO	cluster_19	Broad Commodities	2026-06-03	17	-24.0%	1.00	-1.00
DBC	cluster_19	Broad Commodities	2026-06-03	17	-12.3%	1.00	-1.00
DJP	cluster_19	Broad Commodities	2026-06-03	17	-11.9%	1.00	-1.00
PDBC	cluster_19	Broad Commodities	2026-06-03	17	-12.2%	1.00	-1.00

ETF Net Signal Strength

NSS = P(buy within h=5 days) - P(sell within h=5 days), range [-1, 1]. ETF rows only, cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive trailing return – i.e. would not be blocked by the entry filter.

Top 20 most bullish (cluster-deduped, abs-mom approved)

ticker	cluster	theme	p_buy	nss
PSCH	cluster_11	Healthcare	1.00	1.00
XHB	cluster_1	U.S. Mid Cap Equity	1.00	1.00
EDV	cluster_13	Fixed Income	1.00	1.00
SMDV	cluster_2	U.S. Large Cap Value	1.00	1.00
FXO	cluster_14	Financials	1.00	1.00
HYMB	cluster_16	Municipal Bond	1.00	1.00
PSCC	cluster_18	Consumer Staples	1.00	1.00
IAGG	cluster_6	Aggregate Fixed Income	1.00	1.00
IBUY	cluster_3	U.S. Large Cap Growth	1.00	1.00
CNXT	cluster_12	China Equity	1.00	0.99
USRT	cluster_15	Real Estate	0.99	0.98
MINT	cluster_24	Cash / Ultra-Short Fixed ...	1.00	0.98
FALN	cluster_10	High Yield Bond	0.98	0.98
EWP	cluster_4	Developed ex-US Equity	1.00	0.97
FLTR	cluster_23	Bank Loan / Floating Rate	1.00	0.97
EMLP	cluster_9	Materials	0.80	0.80
LTPZ	cluster_21	Inflation-Protected Fixed...	0.72	0.65
HEWJ	cluster_22	Japan Equity	0.99	0.62
SPHQ	cluster_5	U.S. Large Cap Equity	1.00	0.60
IGF	cluster_8	Asia-Pacific Equity	0.59	0.58

Top 20 most bearish (cluster-deduped)

ticker	cluster	theme	p_sell	nss
DBC	cluster_19	Broad Commodities	1.00	-1.00
NORW	cluster_9	Materials	1.00	-1.00
EWD	cluster_4	Developed ex-US Equity	1.00	-1.00
FXC	cluster_8	Asia-Pacific Equity	1.00	-1.00
KGRN	cluster_12	China Equity	1.00	-1.00
SLV	cluster_17	Gold	1.00	-1.00

PSP	cluster_14	Financials	1.00	-1.00
FXV	cluster_25	International Fixed Income	1.00	-1.00
XLC	cluster_5	U.S. Large Cap Equity	1.00	-1.00
DVYE	cluster_7	Emerging Markets Equity	1.00	-1.00
PFF	cluster_10	High Yield Bond	1.00	-1.00
EW	cluster_20	India Equity	1.00	-1.00
BKLN	cluster_23	Bank Loan / Floating Rate	1.00	-1.00
ACWV	cluster_2	U.S. Large Cap Value	1.00	-1.00
EIS	cluster_3	U.S. Large Cap Growth	1.00	-1.00
EWJV	cluster_22	Japan Equity	1.00	-0.99
BSC	cluster_6	Aggregate Fixed Income	0.98	-0.98
VTIP	cluster_21	Inflation-Protected Fixed...	0.96	-0.96
GSY	cluster_24	Cash / Ultra-Short Fixed ...	0.96	-0.91
QMOM	cluster_1	U.S. Mid Cap Equity	0.87	-0.69

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	compo
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_20	10	FLAT	0.305	0.095	0.00021	
UNG	United States Natural Gas Fund ...	Natural Gas	cluster_19	11	FLAT	0.485	0.076	0.00017	
GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.716	0.041	0.00007	
FXV	Invesco CurrencyShares Japanese...	Currency	cluster_25	6	FLAT	0.730	0.095	0.00022	
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.616	0.218	0.00036	
DBA	Invesco DB Agriculture Fund	Agricultural Commodit...	cluster_19	11	FLAT	0.595	0.171	0.00039	
FTSM	First Trust Enhanced Short Matu...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.756	0.136	0.00032	
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.756	0.162	0.00040	
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.660	0.186	0.00047	
NEAR	iShares Short Duration Bond Act...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.707	0.192	0.00045	
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.734	0.180	0.00044	
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.851	0.025	0.00000	
SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.851	0.083	0.00021	
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_20	10	FLAT	0.501	0.220	0.00056	
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_6	43	LONG	0.846	0.129	0.00036	
CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fixed...	cluster_6	43	LONG	0.876	0.161	0.00031	
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_25	6	FLAT	0.820	0.204	0.00049	
LMBS	First Trust Low Duration Opport...	Mortgage-Backed Fixed...	cluster_6	43	LONG	0.835	0.238	0.00047	
NOBL	ProShares S&P 500 Dividend Aris...	U.S. Dividend Equity	cluster_20	10	FLAT	0.446	0.274	0.00075	
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_16	13	LONG	0.877	0.249	0.00045	
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_20	10	FLAT	0.444	0.329	0.00083	
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_16	13	LONG	0.877	0.264	0.00053	
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_12	15	FLAT	0.799	0.271	0.00071	
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_9	21	FLAT	0.749	0.361	0.00085	
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_20	10	FLAT	0.510	0.358	0.00102	
EPHE	iShares MSCI Philippines ETF	Philippines Equity	cluster_20	10	FLAT	0.564	0.410	0.00107	
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	cluster_23	9	FLAT	0.865	0.326	0.00071	
KBA	KraneShares Bosera MSCI China A...	China Equity	cluster_12	15	FLAT	0.857	0.259	0.00073	
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_12	15	FLAT	0.903	0.241	0.00064	
LTPZ	Pimco 15 Year U.S. TIPS Index E...	Inflation-Protected F...	cluster_21	10	LONG	0.918	0.228	0.00057	

Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

	scope	parameter	value
short_ma	screen	short_ma	5
buy_ma	screen	buy_ma	50
sell_ma	screen	sell_ma	20
vol_window	screen	vol_window	20
horizon	screen	horizon	5
imminent_horizon	screen	imminent_horizon	1
cluster_smoothing_window	screen	cluster_smoothing_window	3
sell_cushion_k	screen	sell_cushion_k	0.5
sell_cushion_sd_window	screen	sell_cushion_sd_window	20